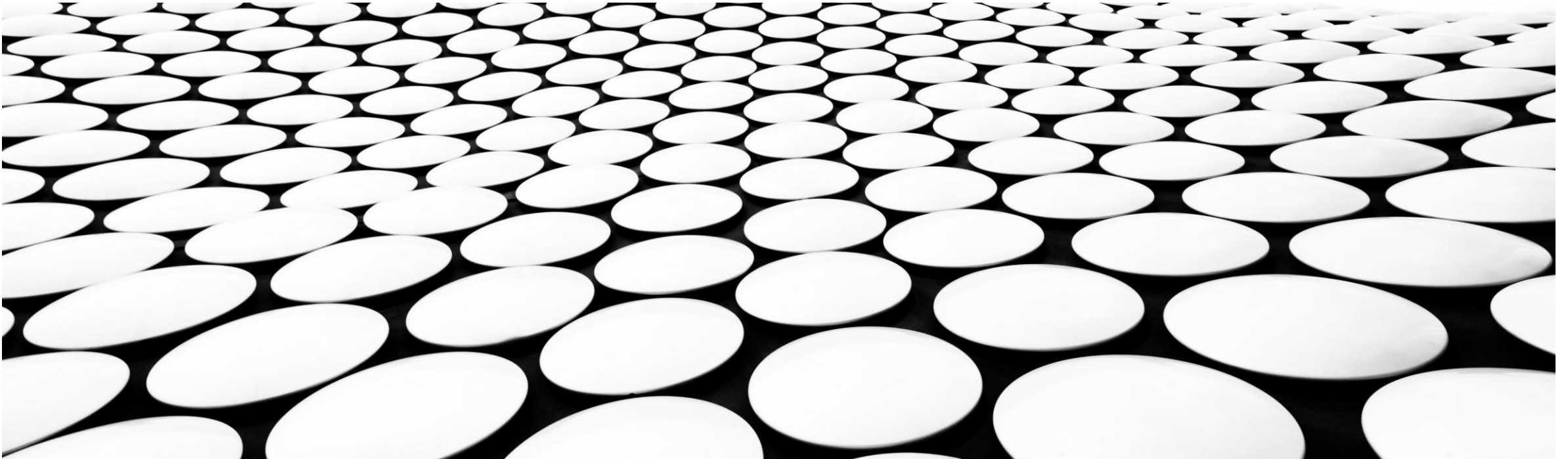
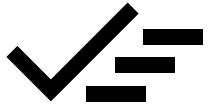


TIME SERIES ANALYSIS

Segmentation, Classification, Forecasting





TIME SERIES ANALYSIS

- Introduction and Applications
- Segmentation
- Classification
- Forecasting

Time Series

Introduction and Applications

Definition

Time Traveling with Data Science: Focusing on Time Series Analysis (Part 1)

Digitale Transformation, Künstliche Intelligenz 20. Juli. 2021

Dr. Julien Siebert, Christof Schroth und Janek Groß

[Source](#)



Blog des Fraunhofer-Institut für
Experimentelles Software Engineering

A time series is an ordered set of data points generated from successive measurements over time.

For example, in the context of IoT, measurements from sensors typically comprise time series (e.g., physical values such as pressure, temperature, electrical voltage, etc.).

Data scientists have a wide range of tools at their disposal to analyze time series. Some of these techniques are developed in different contexts (bioinformatics, neuroscience, signal processing, economics, etc.) and implemented in various libraries.

Characteristics

Main focus in this course:
univariate + regular

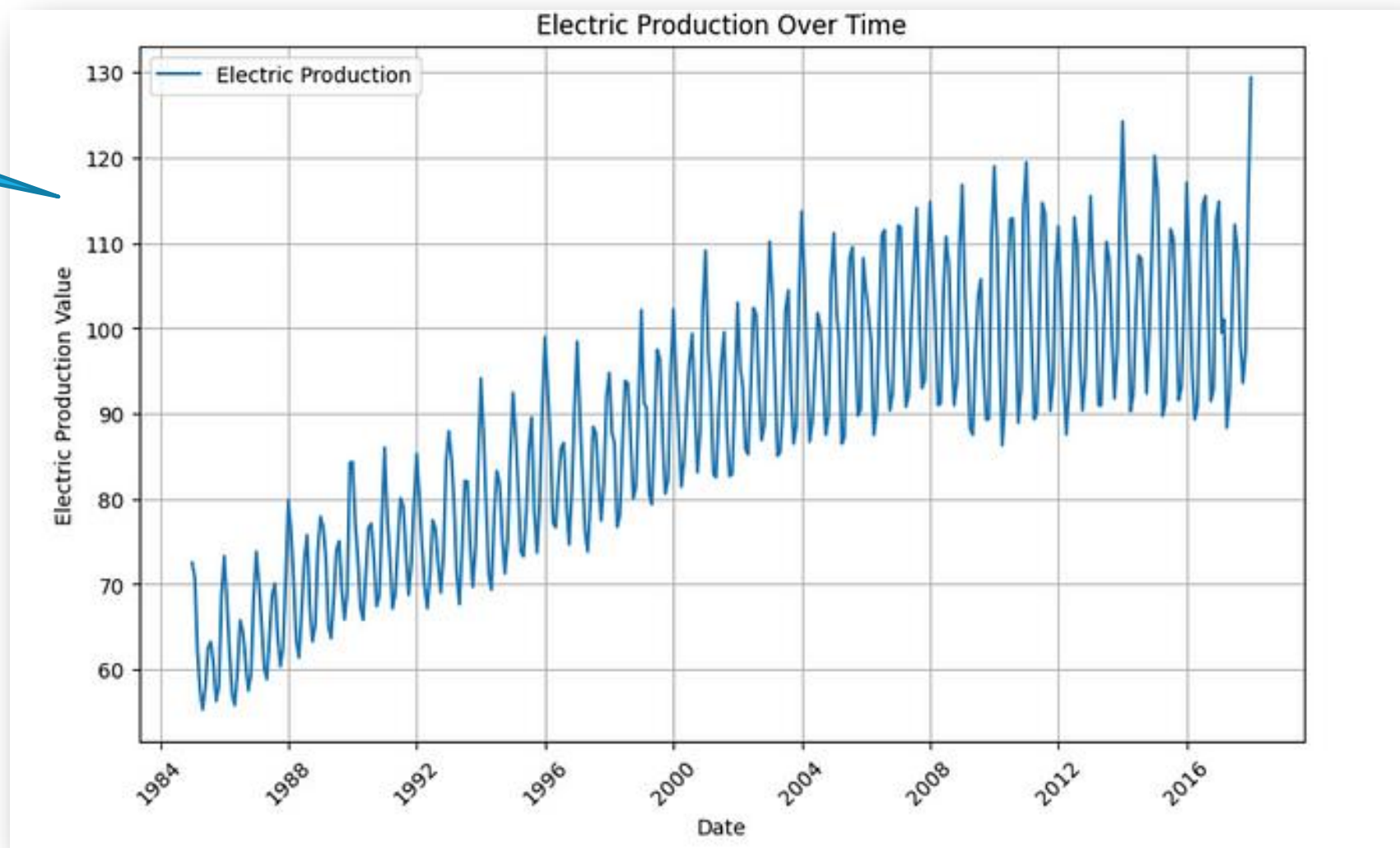
Univariate vs. Multivariate Time Series:

- Univariate: A single variable or feature recorded over time (e.g., daily temperature).
- Multivariate: Multiple variables recorded over time (e.g., daily temperature, humidity, and wind speed).

Regular vs. Irregular Time Series:

- Regular: Data points are recorded at consistent time intervals (e.g., hourly, daily).
- Irregular: Data points are recorded at inconsistent time intervals.

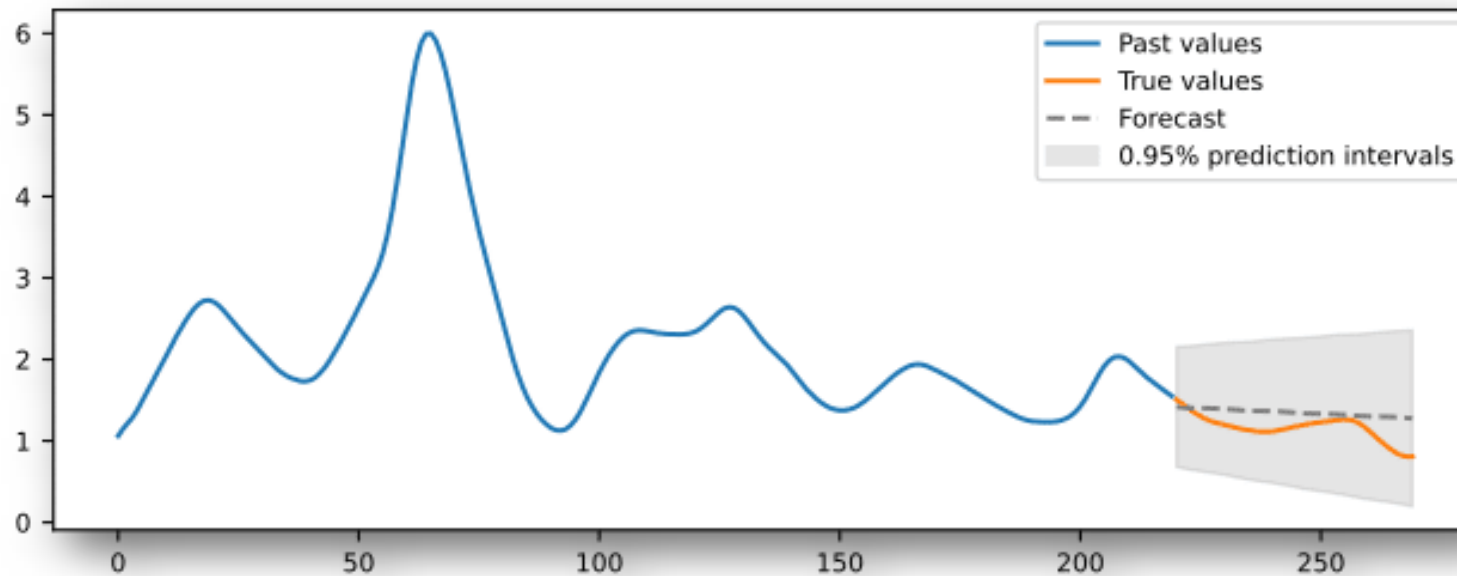
Regular univariate time series



Types of analysis

- Forecasting
- Indexing
- Clustering
- Classification
- Anomaly Detection
- Segmentation
- Blind Source Detection

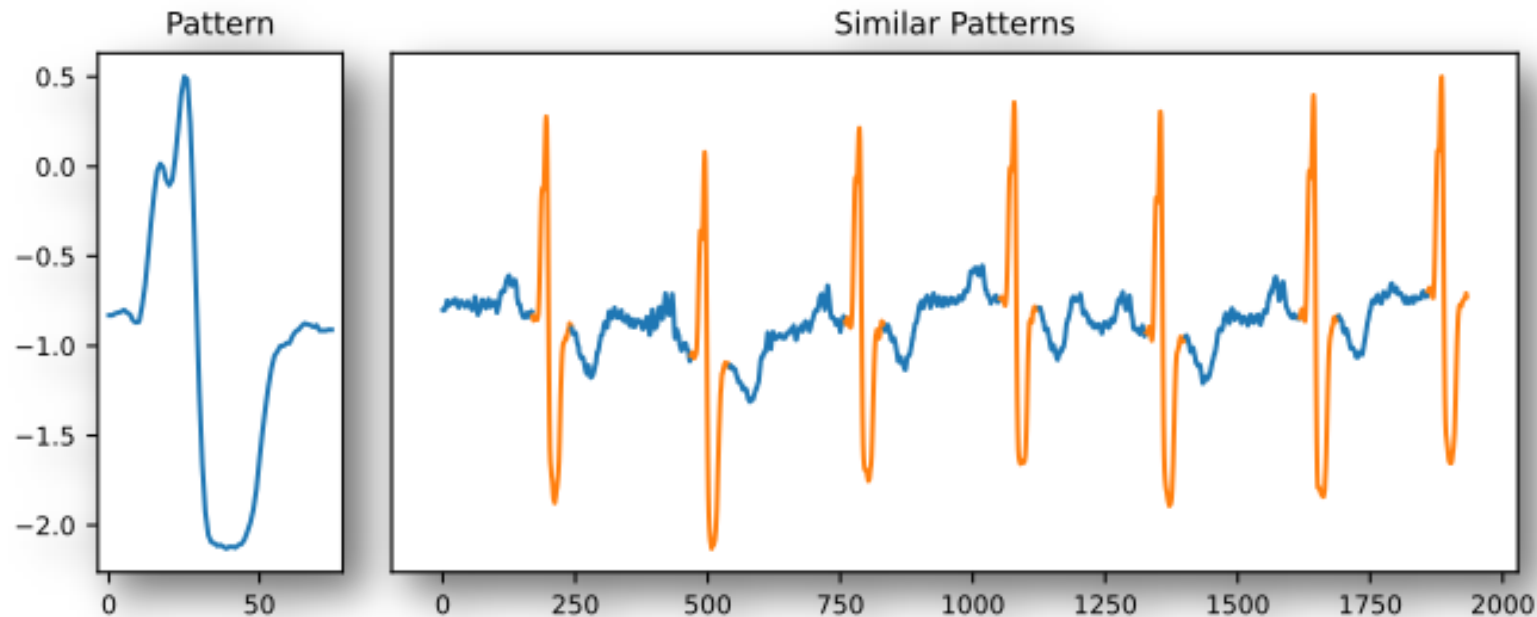
Forecasting (also called **prediction**) is the task of predicting future values of a time series given some past data. This is probably the best-known and most frequently used task.



Examples

- Predict electricity demand for power grid management.
- Forecast blood sugar levels for diabetic patients.

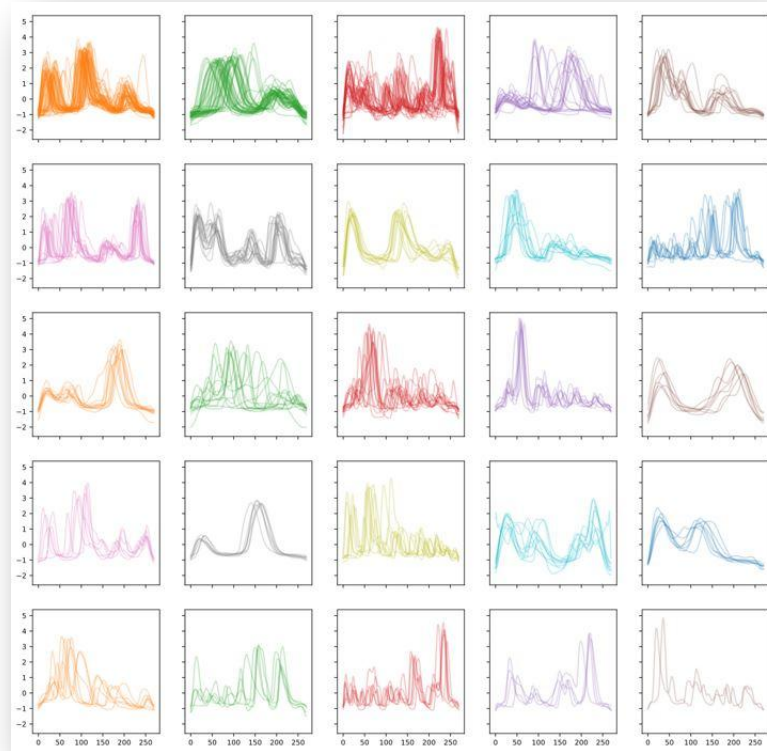
Indexing (also called **query by content**) is the task of finding time series in a database containing a particular pattern or subsequence.



Examples

- Search historical machine sensor logs to match an unusual vibration pattern found in a machine (failure).
- Search past ECG segments from a large database to find ECGs containing the same abnormal heartbeat pattern observed in a new patient.

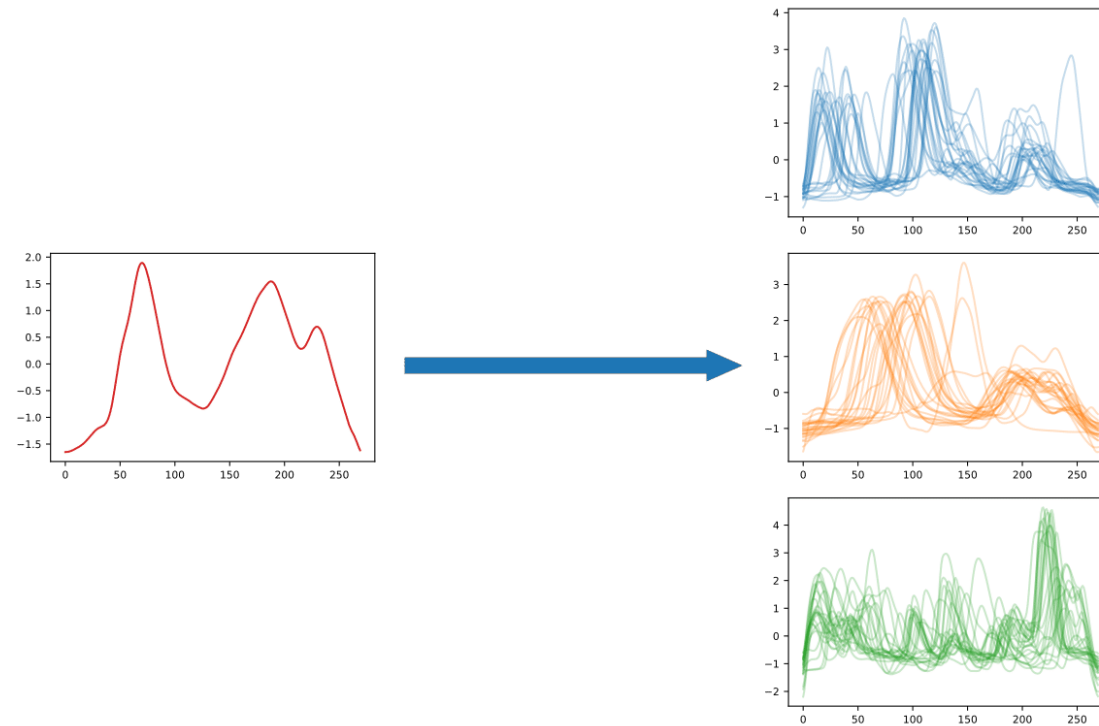
Clustering is the task of grouping similar time series into clusters.



Examples

- Group different patterns of electricity consumption among customers to better understand them.
- Group stocks based on similar price movement patterns over time.

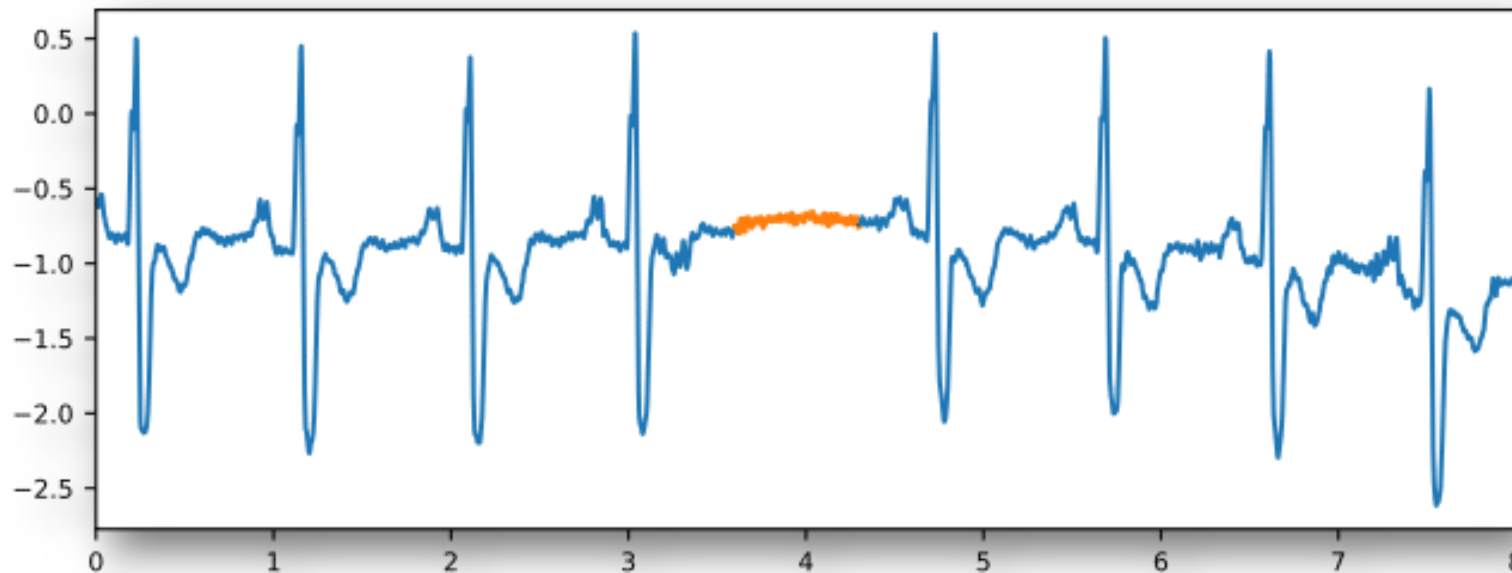
Classification is the task of assigning a predefined class to a time series.



Examples

- Classify speech recordings into emotions like happy, sad, angry, neutral.
- Detect whether an ECG signal represents a normal heartbeat or an arrhythmia.

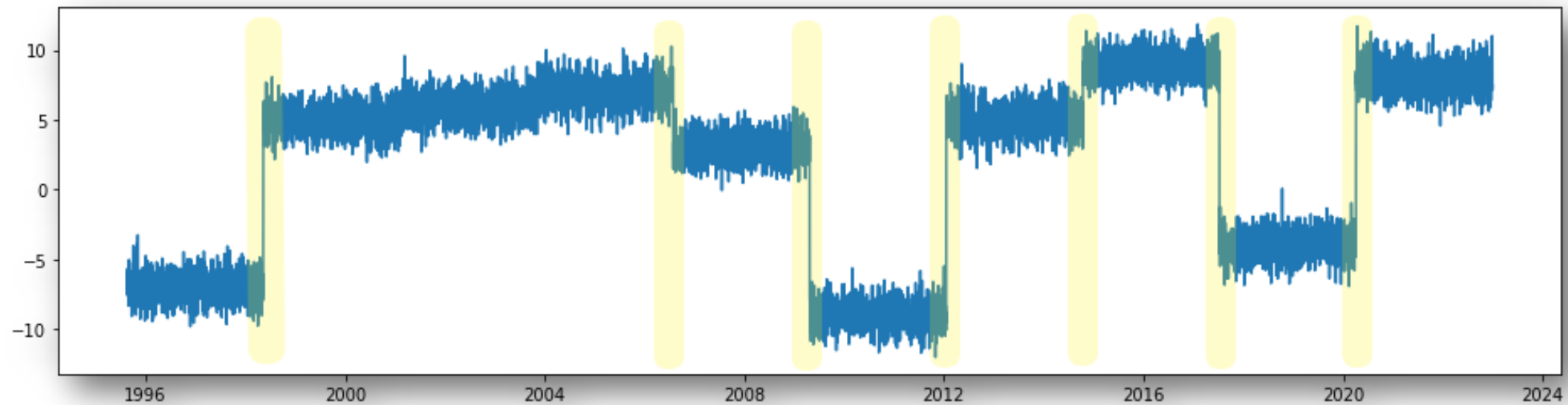
Anomaly detection (also called **outlier detection** or **novelty detection**) is the task of finding abnormal data points (called outliers) or subsequences (called discords).



Examples

- Detect anomalous bank transactions that may indicate fraud.
- Detect abnormal points (or short sequence) in vital signs (e.g., heart rate, blood pressure) that could signal a medical emergency.

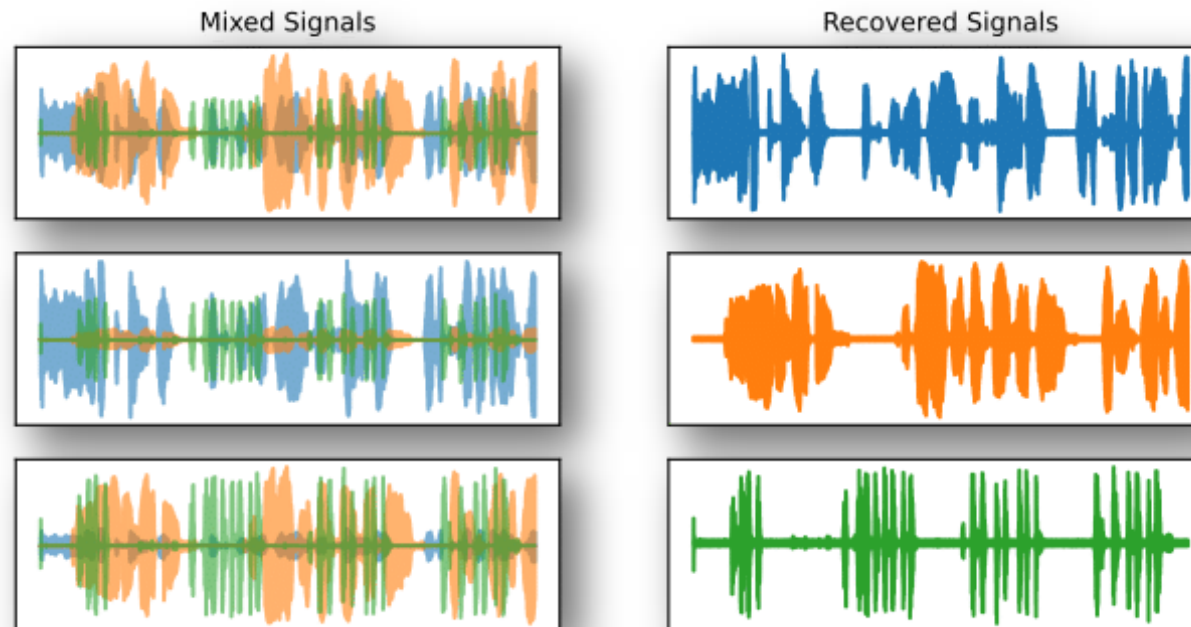
Segmentation of a time series divides long sequences into smaller, meaningful segments.



Examples

- Identify changes in real-time traffic flow data to understand the different phases (e.g., rush hour vs. low traffic periods).
- Identify changes in mention frequency that could point to a crisis event or public relations incident.

Blind source separation is the task of recovering the source signals from a set of mixed signals. The challenge is to do this with very little information about the sources themselves or the mixing process.

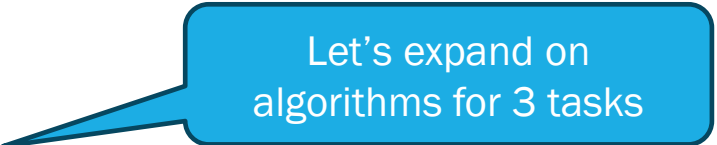


Examples

- Separate different people's voices from a recording of a noisy room (“Cocktail party problem”).
- Isolate earthquake signals from background noise in seismic sensor data.

Different types of analysis

- Forecasting
- Indexing
- Clustering
- Classification
- Anomaly Detection
- Segmentation
- Blind Source Detection

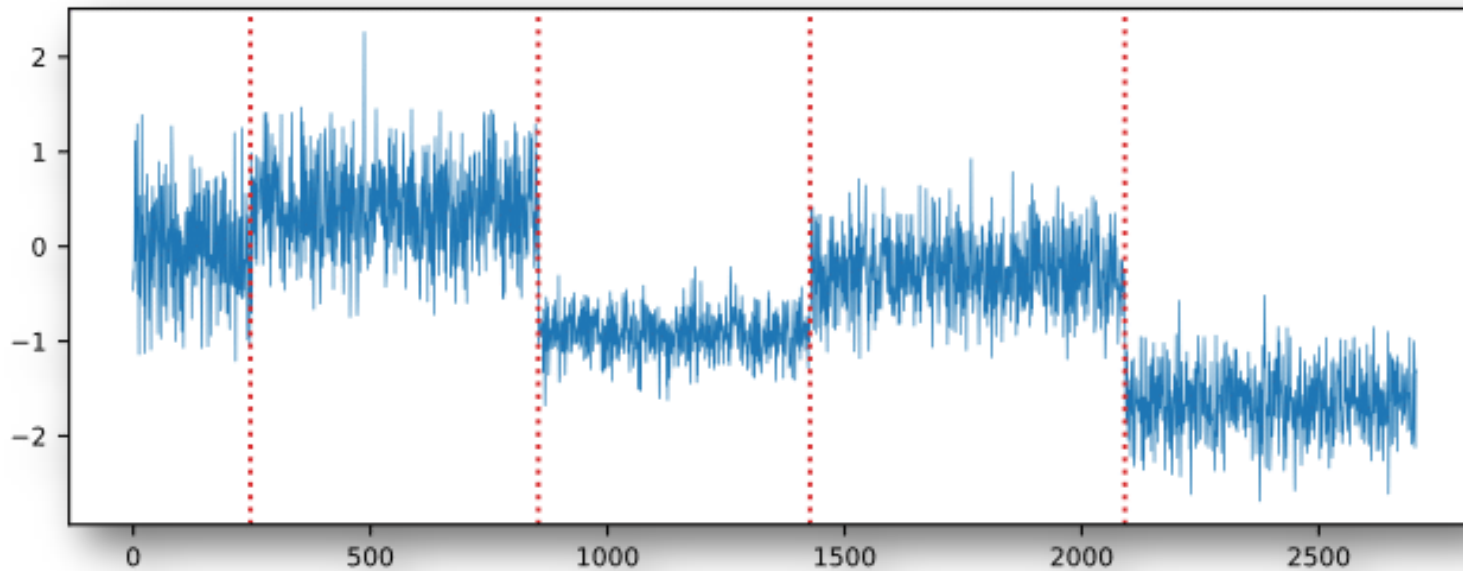


Let's expand on
algorithms for 3 tasks

Segmentation

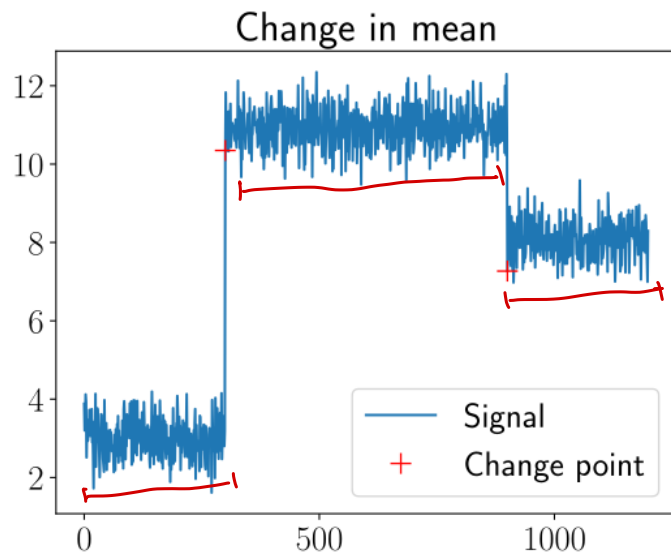
Time series segmentation allows us to divide long sequences into smaller, meaningful segments.

Change point detection is the task of finding points in time where the statistical properties of the time series (like mean, variance) change abruptly.



Common algorithm for
segmentation

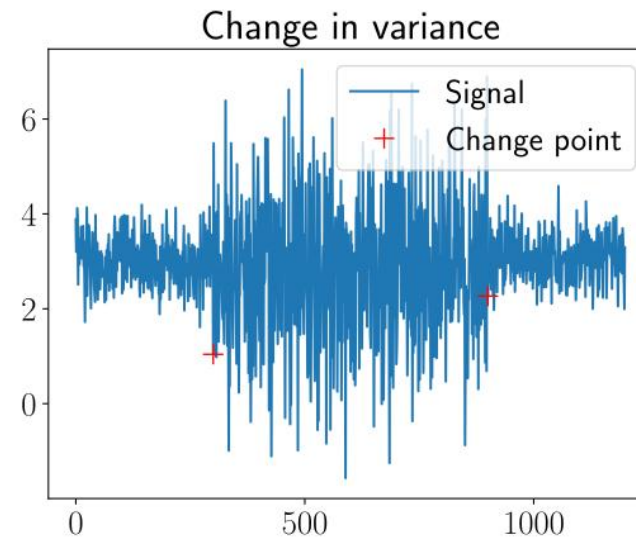
Change in mean occurs when a time series can be divided into different constant segments having different mean values.



moving window

calculate average at different windows

Change in variance could have a stable mean, but there are several segments with different variance values. This can be interpreted as a sudden noise in the signal.



moving window

Both change in mean and change in variance can be detected by comparing statistical properties through the signal.

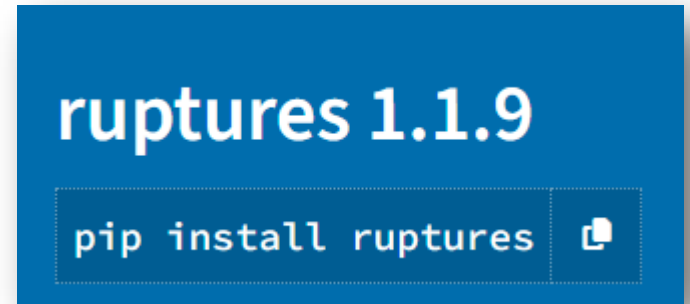
Moving window

A common way to conduct change point detection is a sliding window through the signal. The main idea is to walk through the signal with a window of fixed size.

For each step, *a cost evaluation is done related to the chance of having a change point in the current window.*

Usually, the costs are “low” as long as there is no change in the window and “high” if there is a change in the window.

Pre-defined thresholds can be set for detection.

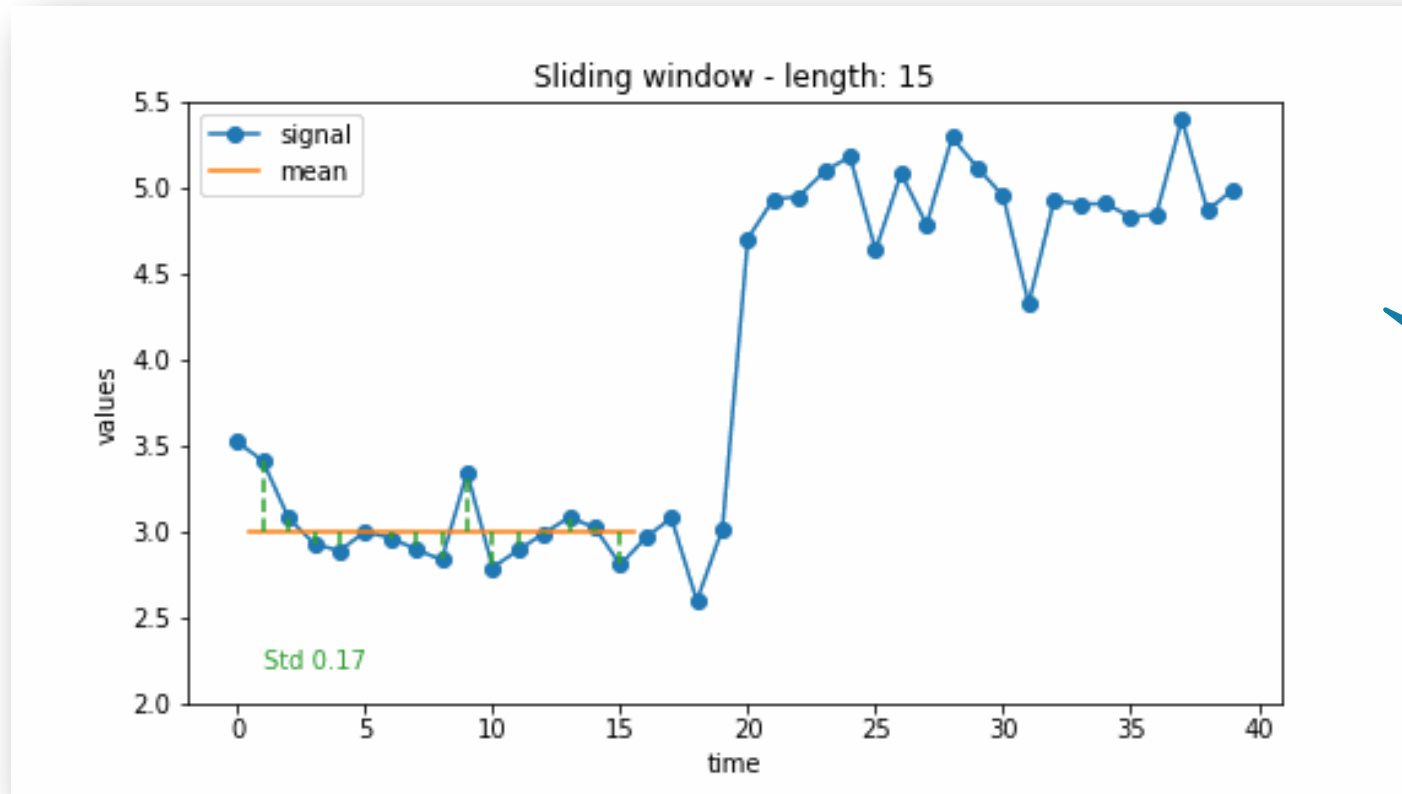


[Source](#)

Packages like ruptures
exist for such CPD tasks

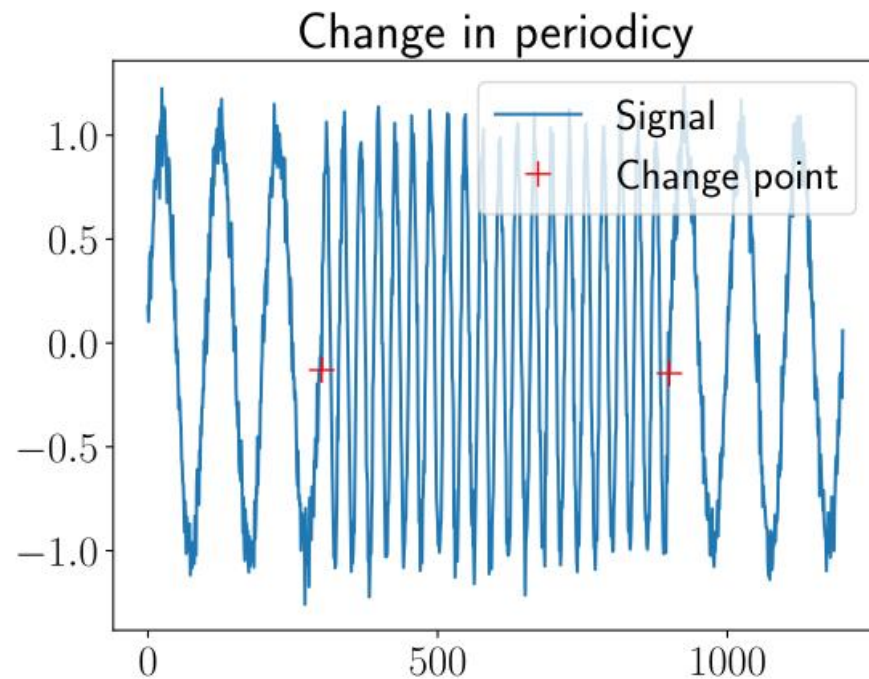
There are packages for time series

To detect changes in mean, a simple approach is to *use the standard deviation as a cost function*. As long as the signal is constant (with some noise), the standard deviation is low. If there is a jump in the signal, the standard deviation will rise.



The sliding window has a particular size

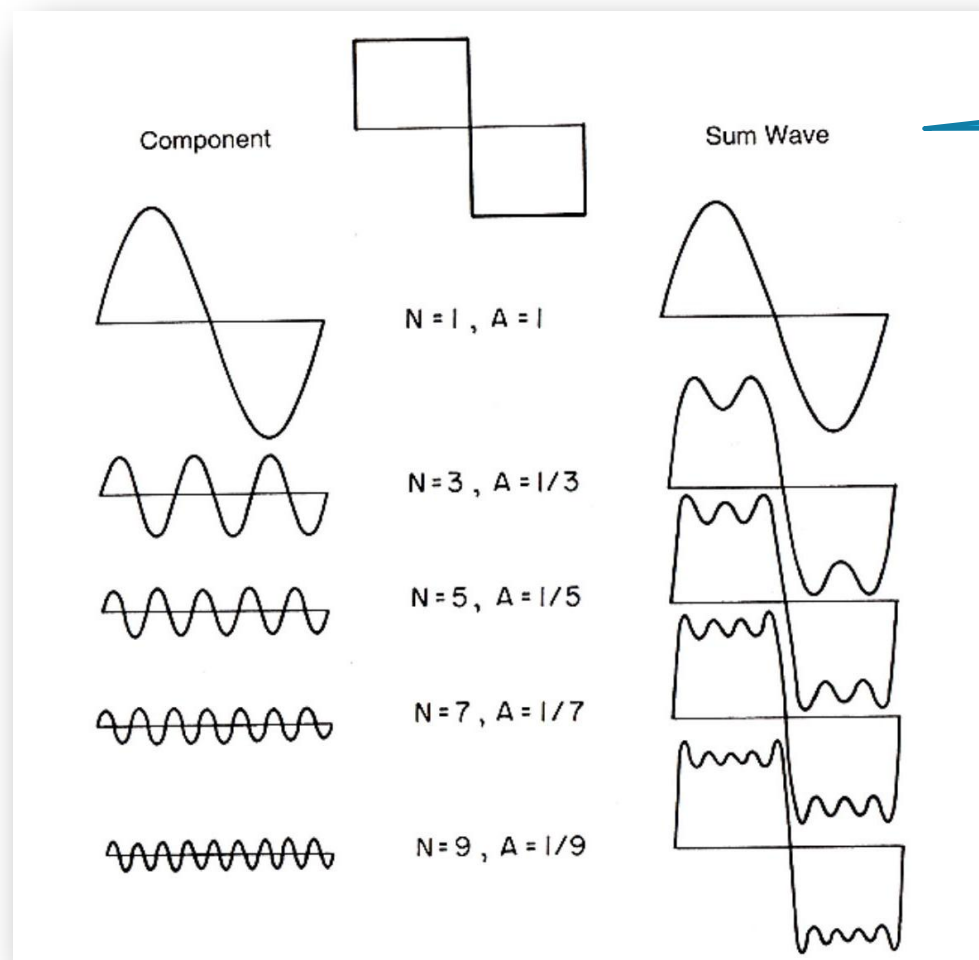
Change in periodicity (also called change in frequency) concerns time series with cyclic properties (e.g., a machine's regime). Here, the change occurs when the frequency changes suddenly.



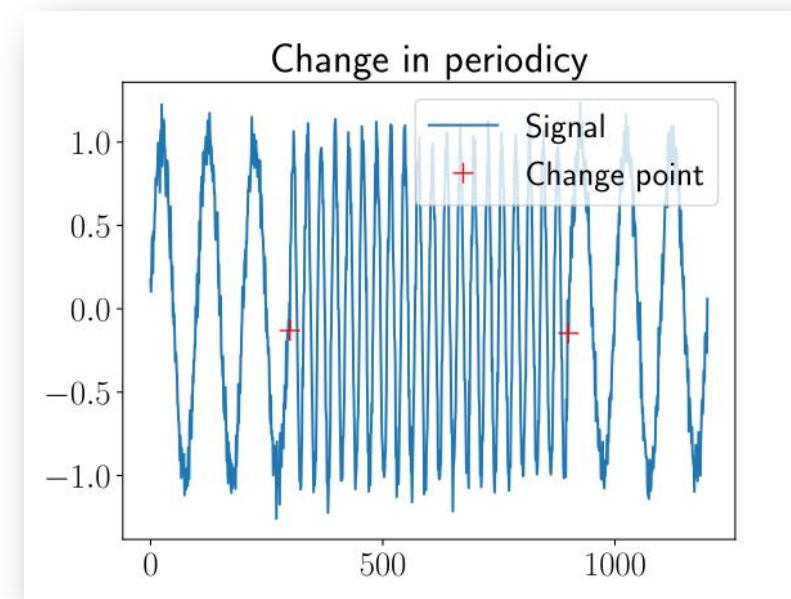
Use of mean/variance is not appropriate here.

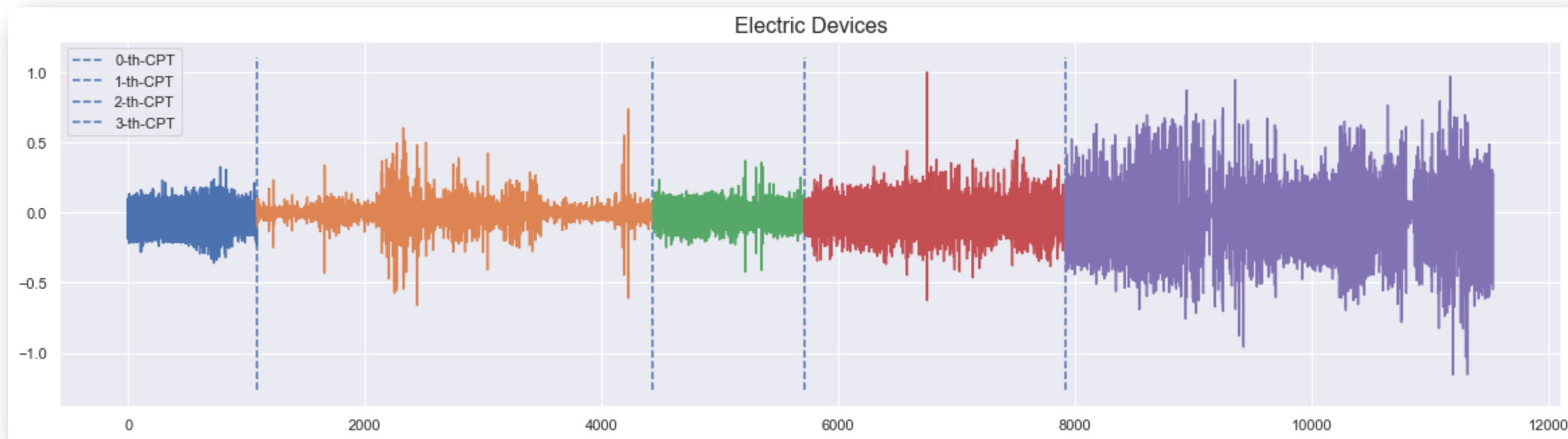
Detection of this kind of change is usually done in the frequency domain, for example by using Fourier transform or wavelet transform.

Intuition of the Fourier Transform:



Fundamental frequency





Trends in Ecology & Evolution



Volume 36, Issue 3, March 2021, Pages 196-205

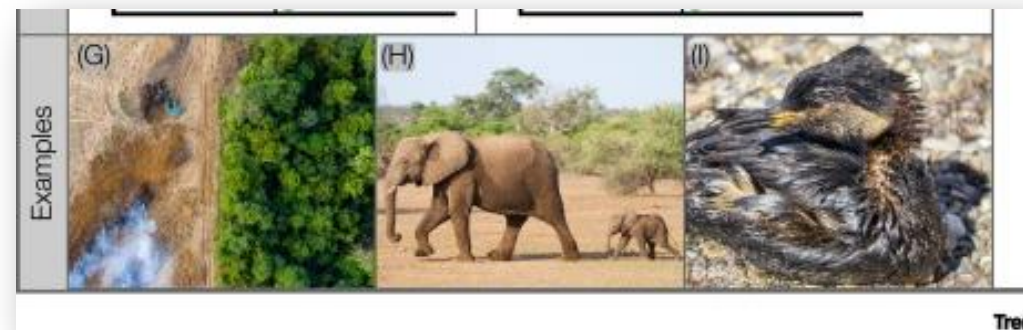
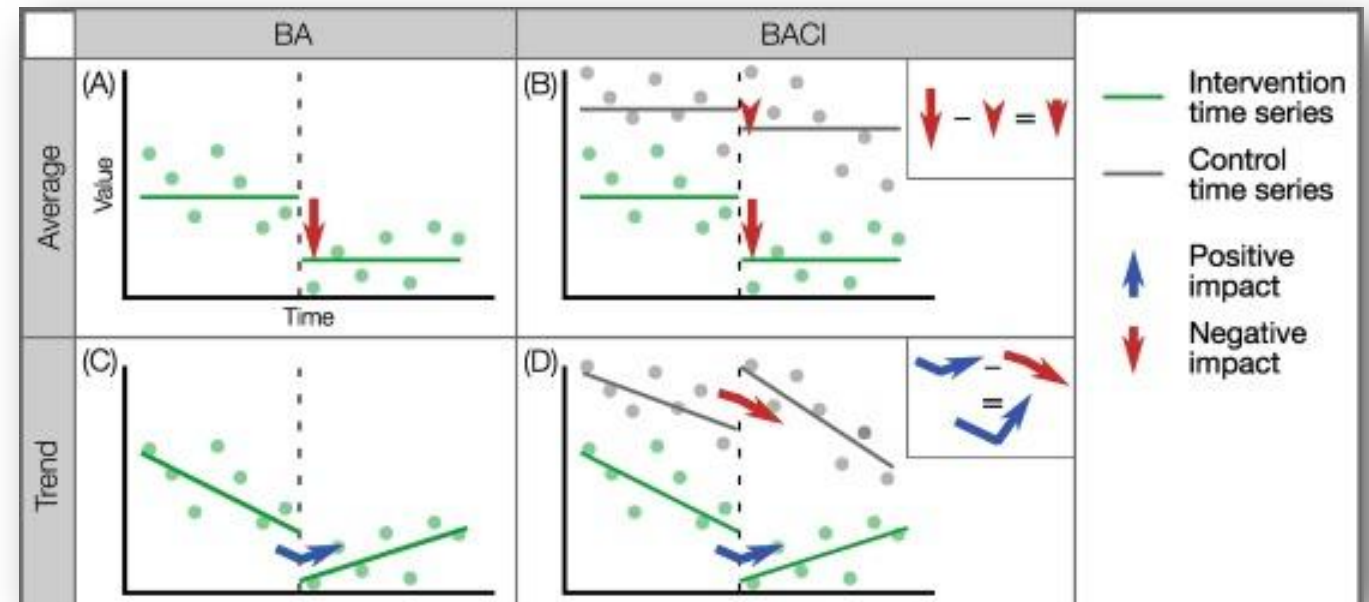
Opinion

Evaluating Impact Using Time-Series Data

Hannah S. Wauchop^{1,2}, Tatsuya Amano^{3,4}, Jonas Geldmann^{1,5}, Alison Johnston^{1,6},
Benno I. Simmons^{1,2,7}, William J. Sutherland¹, Julia P.G. Jones⁸

Source

In intervention, we
« hopefully » create a
segmentation point



Classification

A Guide to Time-Series Sensor Data Classification Using UCI HAR Data

Using TS-Fresh, scikit-learn, and Data Studio to classify sensor data



Chris Knorowski · [Follow](#)

Published in TDS Archive · 8 min read · Jan 27, 2025

[Source](#)

Example of a classification project

4 classes

The *BasicMotions* dataset, donated by Jack Clements to the UEA archive and publicly accessible through the GPL license.

This dataset compiles data from four students wearing a smartwatch and performing different activities: **standing, walking, running and playing badminton**.

6 features

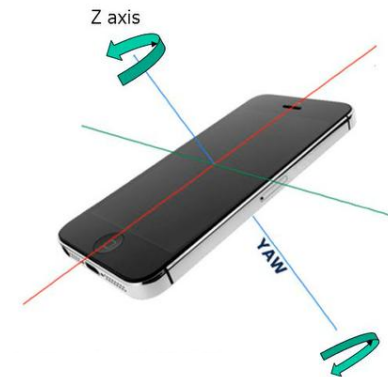
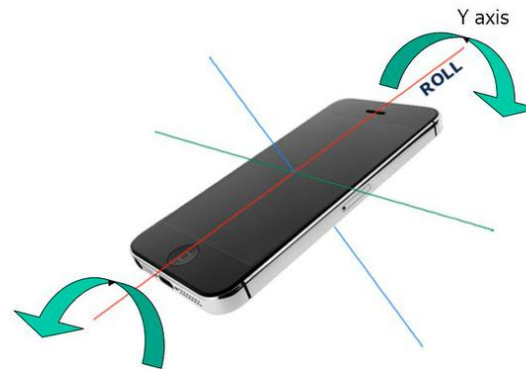
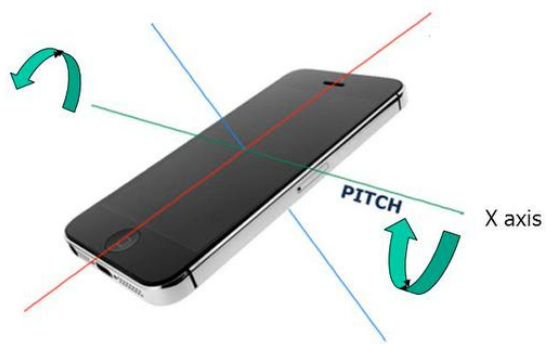
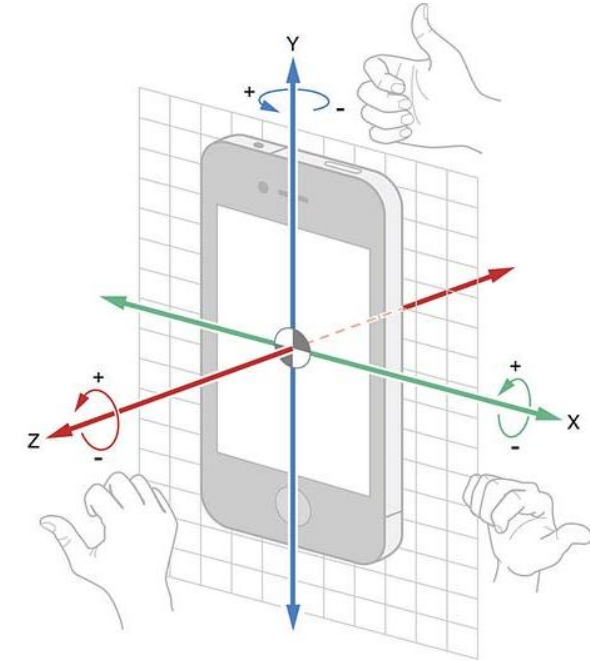
The watch has an **accelerometer** and a **gyroscope** that recorded data in three different axes (**x, y, z**), resulting in six features in total.

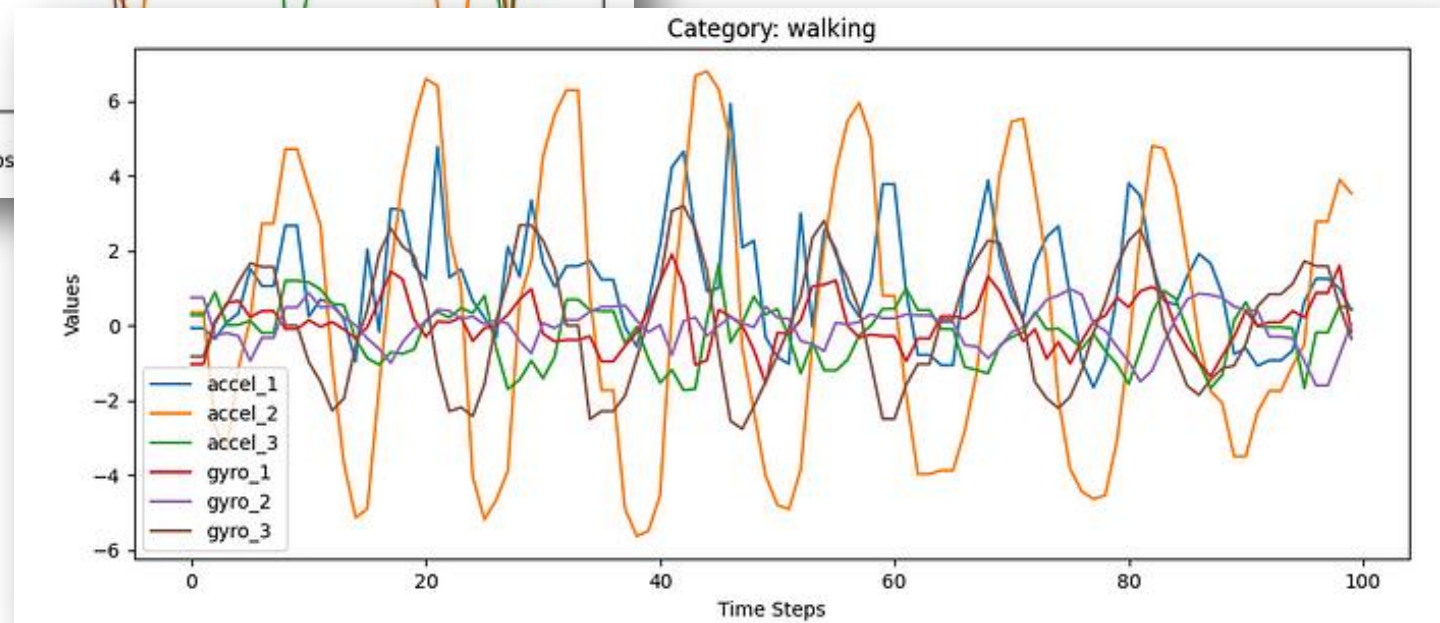
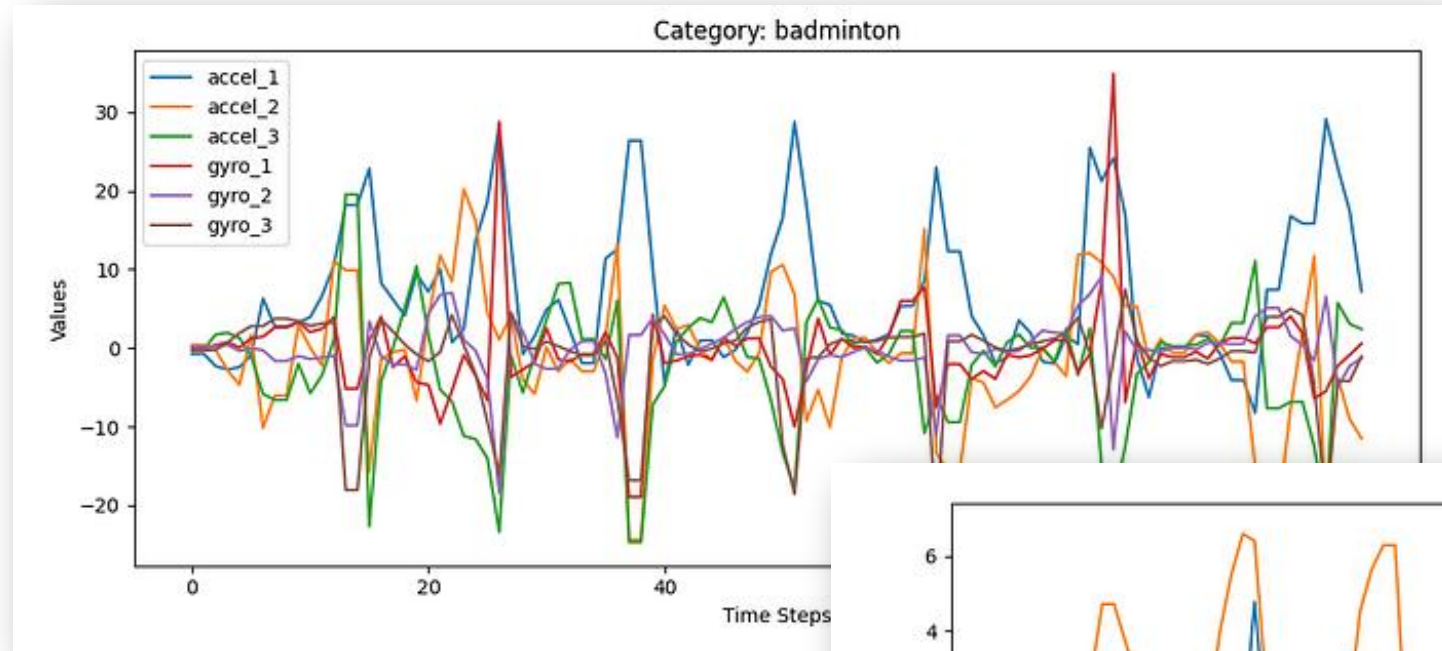
However, the dataset is rather small with only 40 training and testing samples.

Very small dataset

Accelerometers are electromechanical devices that measure acceleration (the rate of change in velocity of an object).

Gyroscope is a device used for measuring rotational changes or maintaining orientation.





Classification models

1. Distance-based (KNN with dynamic time warping)
2. Interval-based (Time Series Random Forests)



2 very different
approaches

Distance-based Models

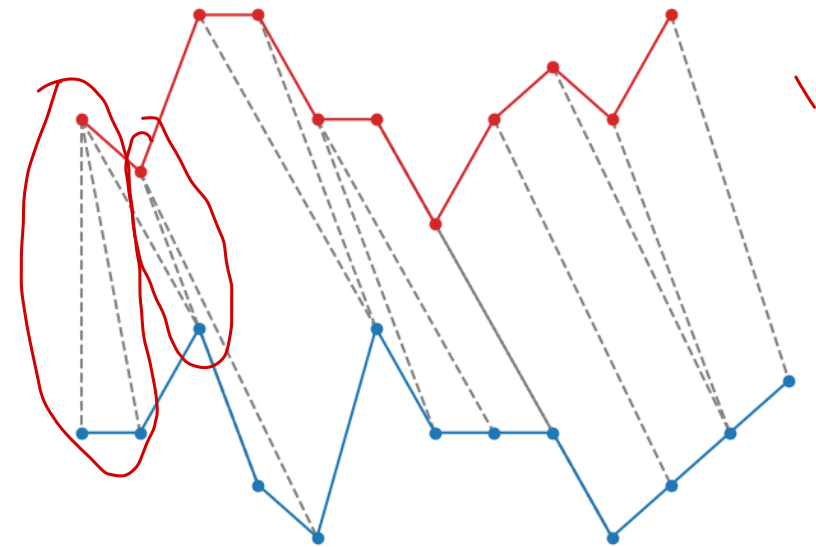
These models rely on a distance metric to classify samples.

We can use kNearestNeighbour (lazy learner), with any distance measure.

For time series, Dynamic time warping (DTW) is a more robust distance measure than Euclidian Distance.

DTW finds the optimal match between each point of two series, allowing it to handle series of different lengths and recognize patterns that are slightly out of phase.

Example of DTW



A brief introduction to Distance Measures

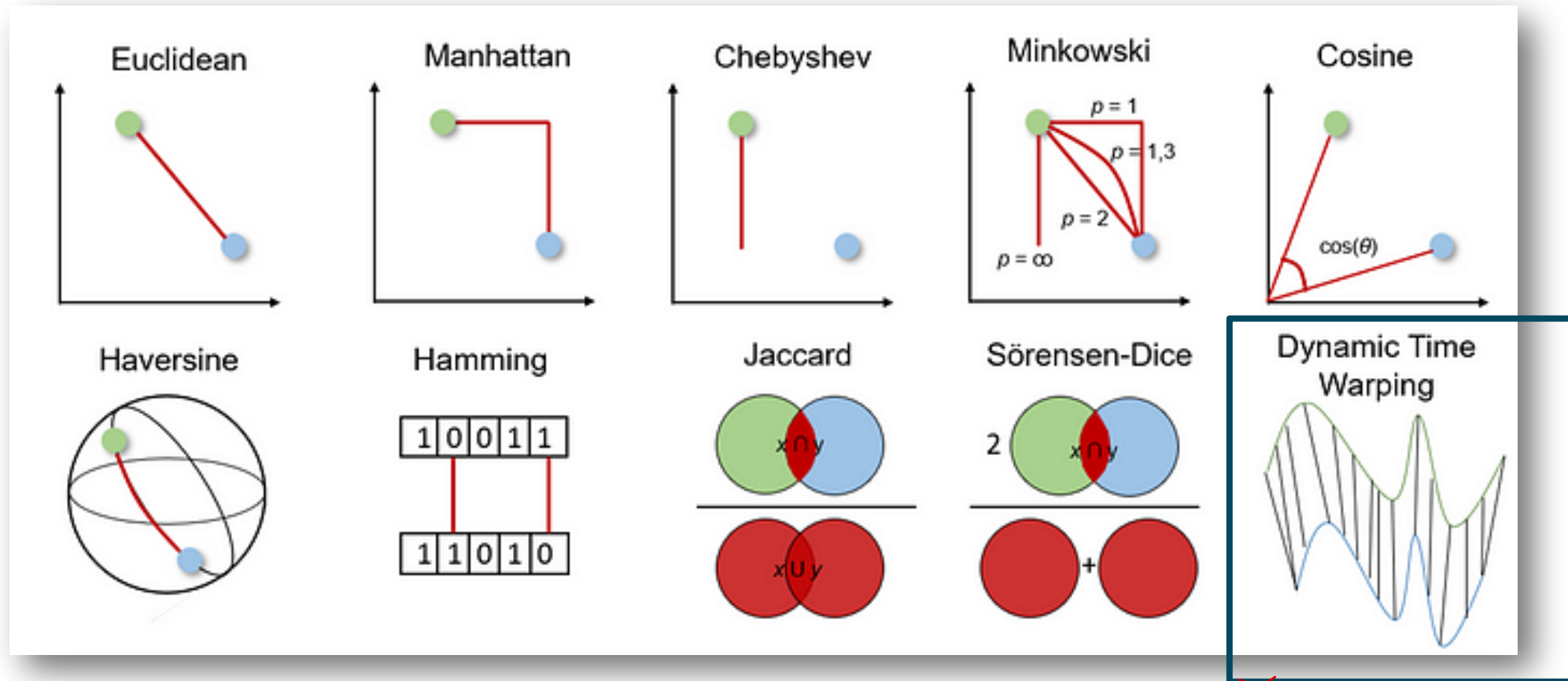
10 distance measures for machine learning you should have heard of



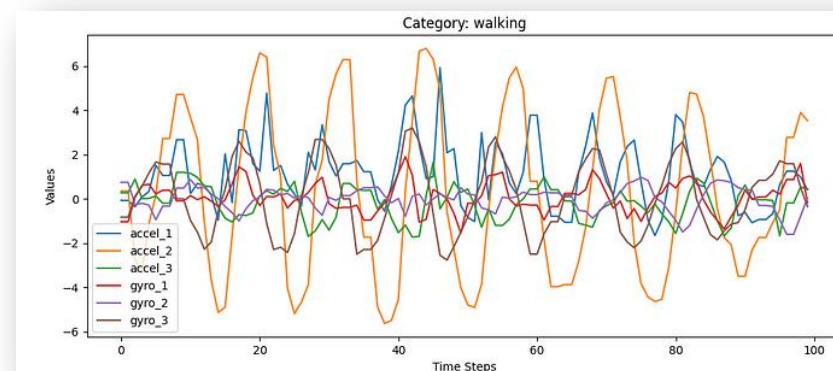
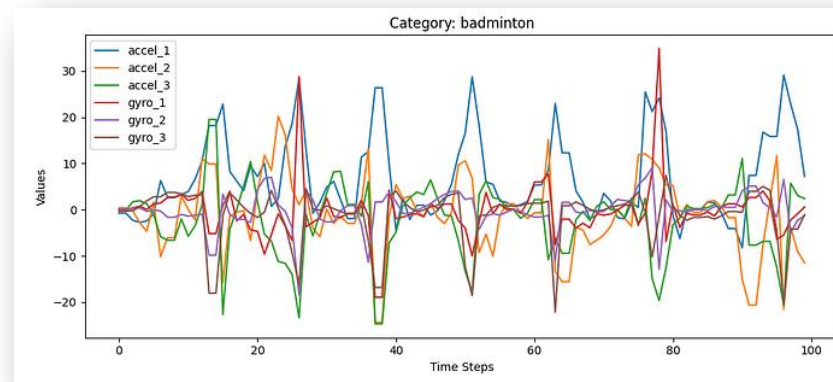
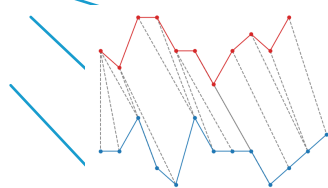
Jonte Dancker · Follow
9 min read · Oct 25, 2022

[Source](#)

Remember this slide?



New time series (all 6 measures)

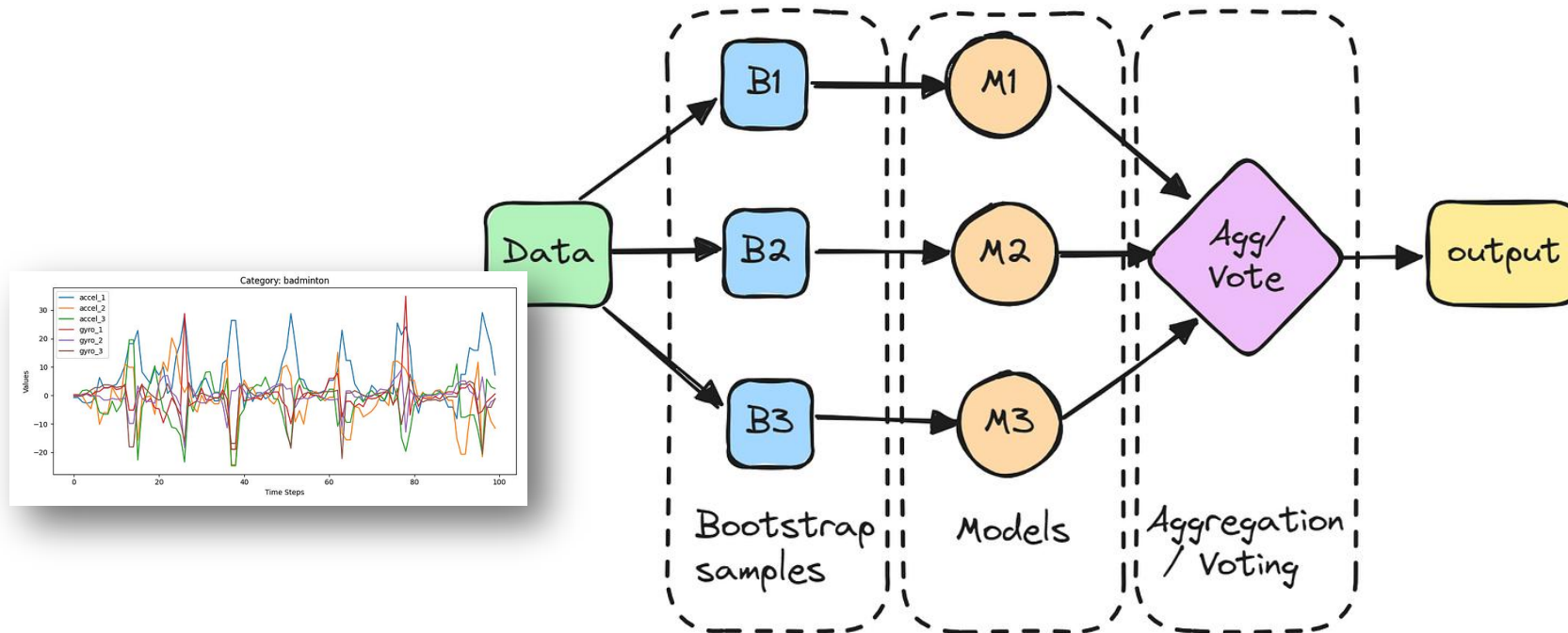


Time series for Standing

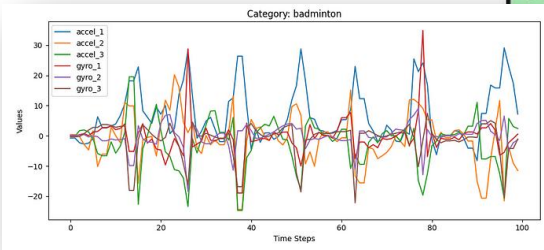
Time series for Running

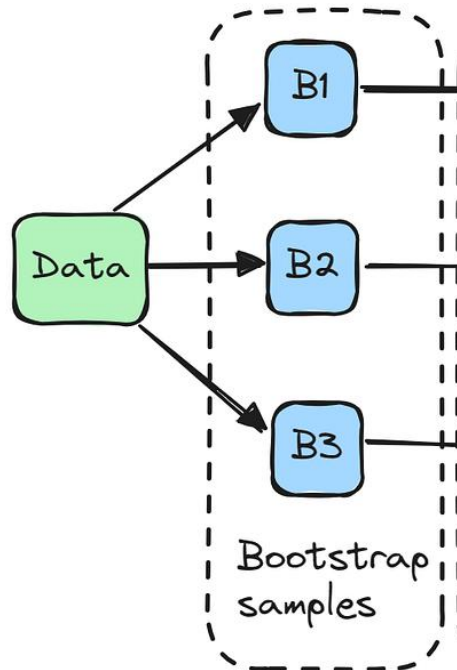
Interval-Based Models

Time Series Random Forest (Bagging approach)



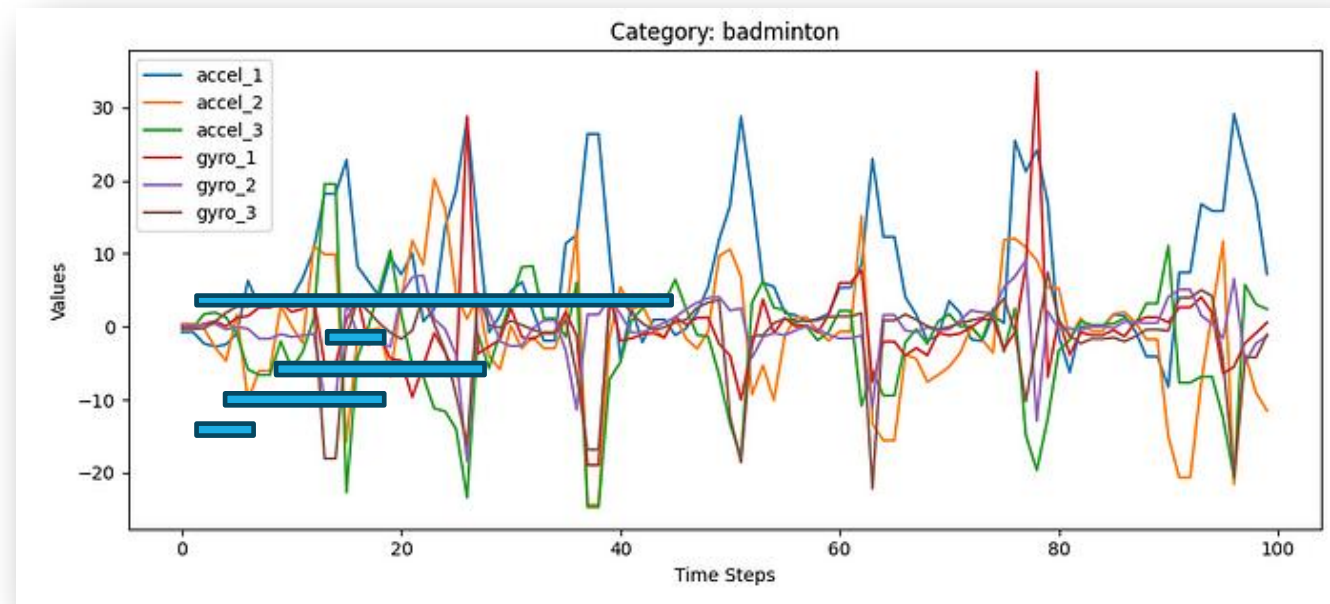
1. Bootstrap Sampling
2. Train Models
3. Aggregate Predictions

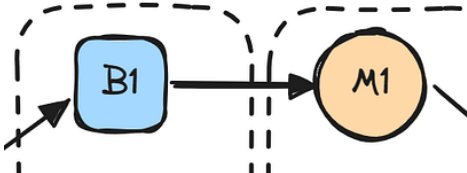
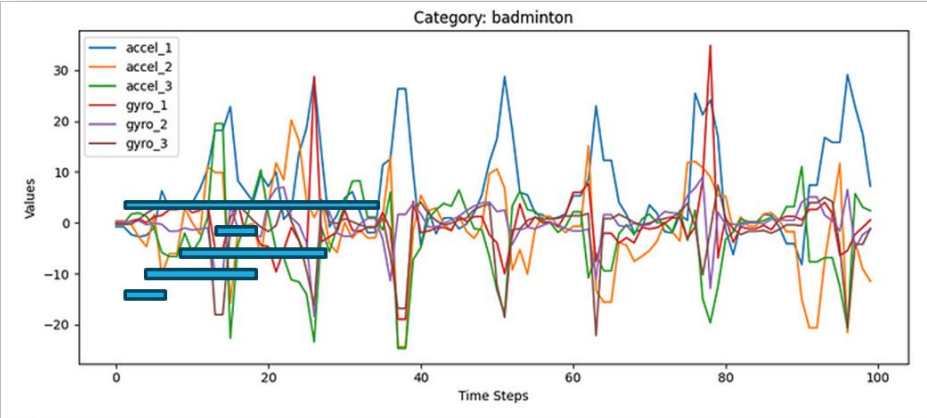
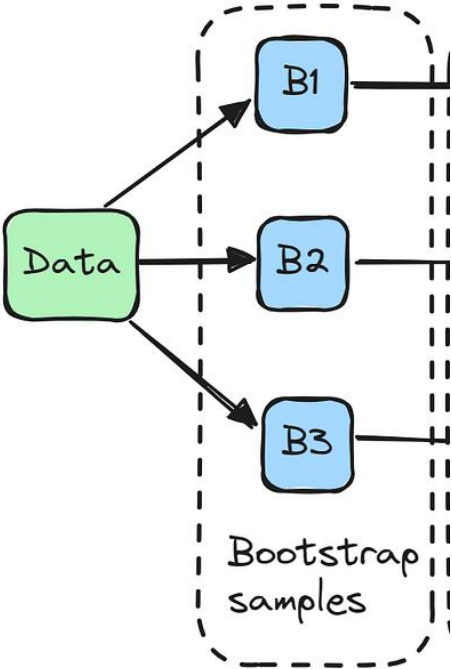




Bootstrap Sampling (for each model – e.g. B1)

- Split the series into N random intervals, with random start positions and random lengths.
- Extract summary features (mean, standard deviation, slope or other features) from each interval into a single feature vector.
- Use the N feature vectors as training set for B1.



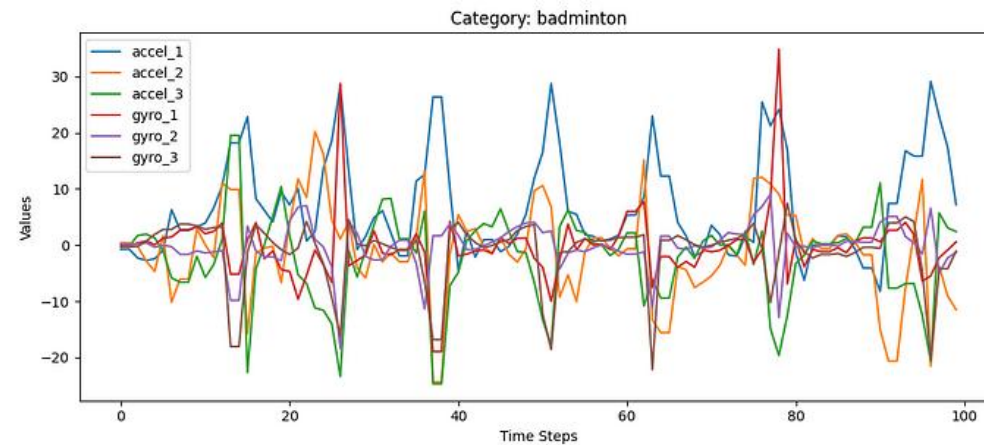
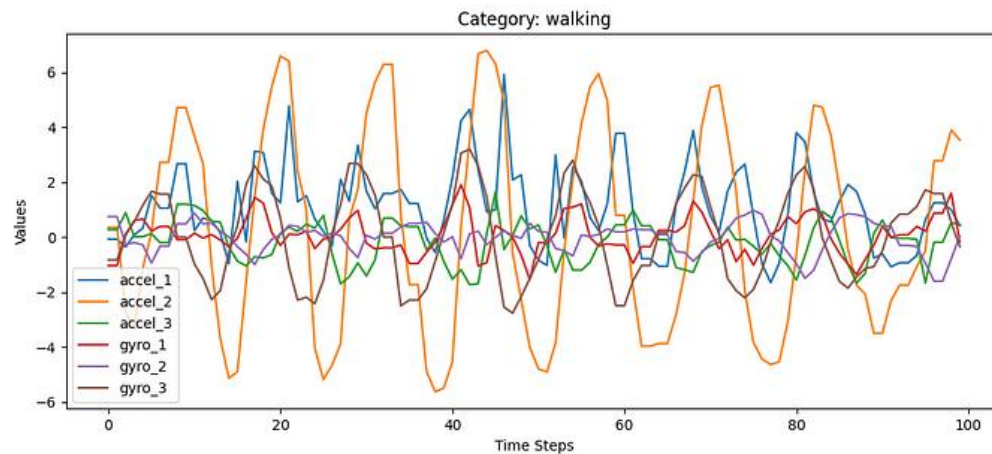


Interval	Signal	Length	Mean	Std	Class
1	Accel_1	30	10	5	Badminton
2	Accel_1	10	5	2	Badminton
					Walking
					Running

Classification models

1. Distance-based (KNN with dynamic time warping)
2. Interval-based (Time Series Random Forests)

We would need to train/test models to evaluate them



Forecasting

Basic models

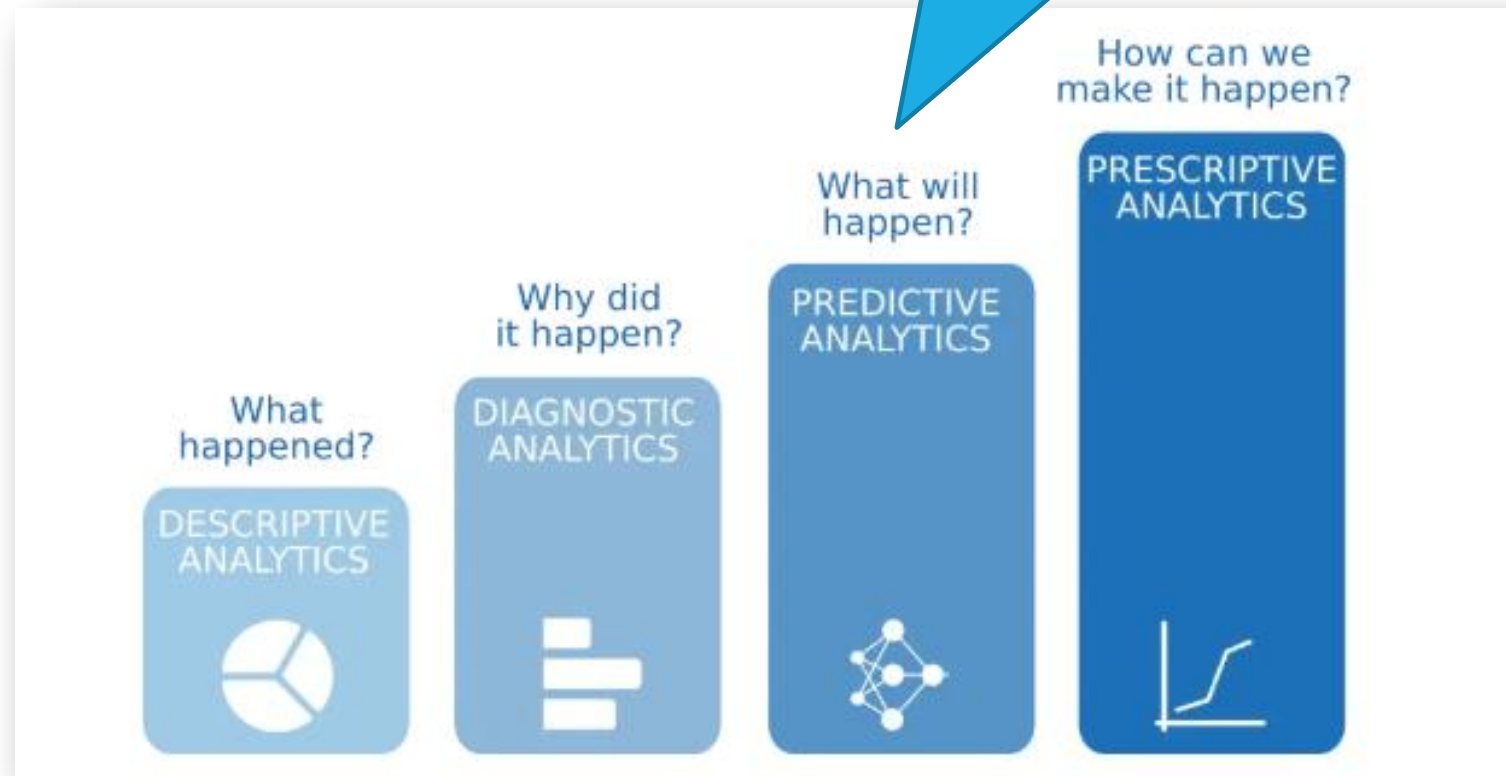
What is advanced analytics?

Tutorials

Pablo Martín

August 31, 2023

A term for the most common task is
FORECASTING

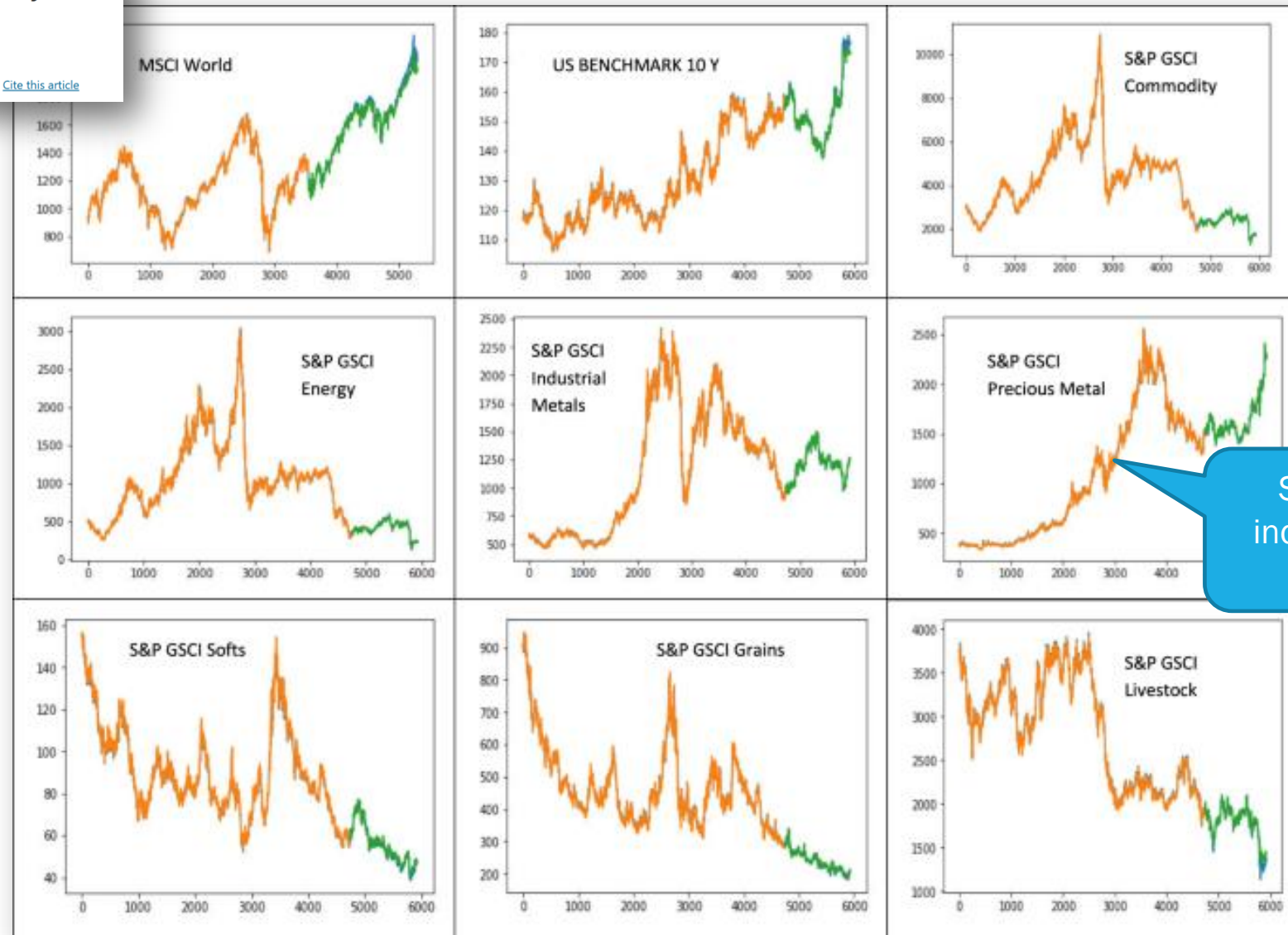


Financial time series prediction under Covid-19 pandemic crisis with Long Short-Term Memory (LSTM) network

Mourad Mroua & Ahlem Lamine

Humanities and Social Sciences Communications 10, Article number: 530 (2023) | [Cite this article](#)

Source



Stock market,
individual stocks,
stock index

Time Series Analysis & Climate Change

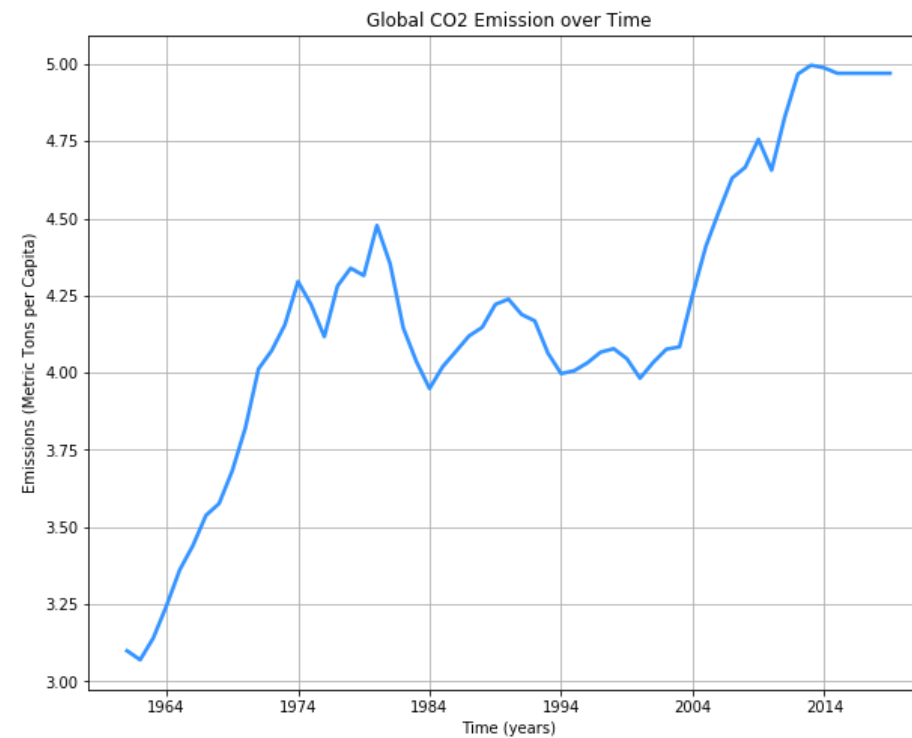
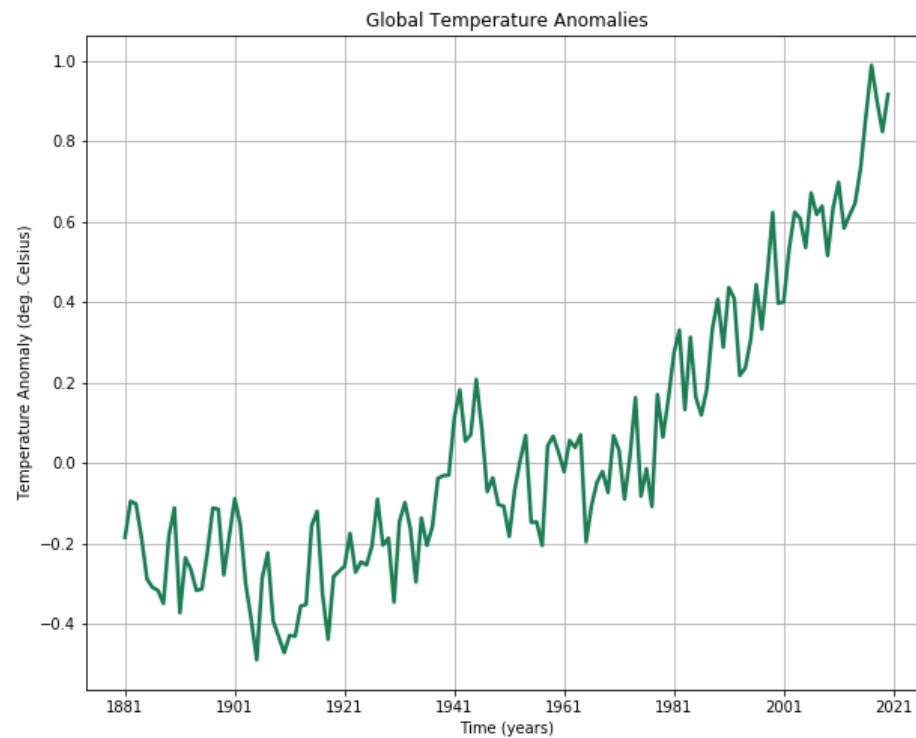
An introductory, hands-on guide to time series analysis and forecasting; investigating climate data using Python, Pandas, and Facebook's Prophet library

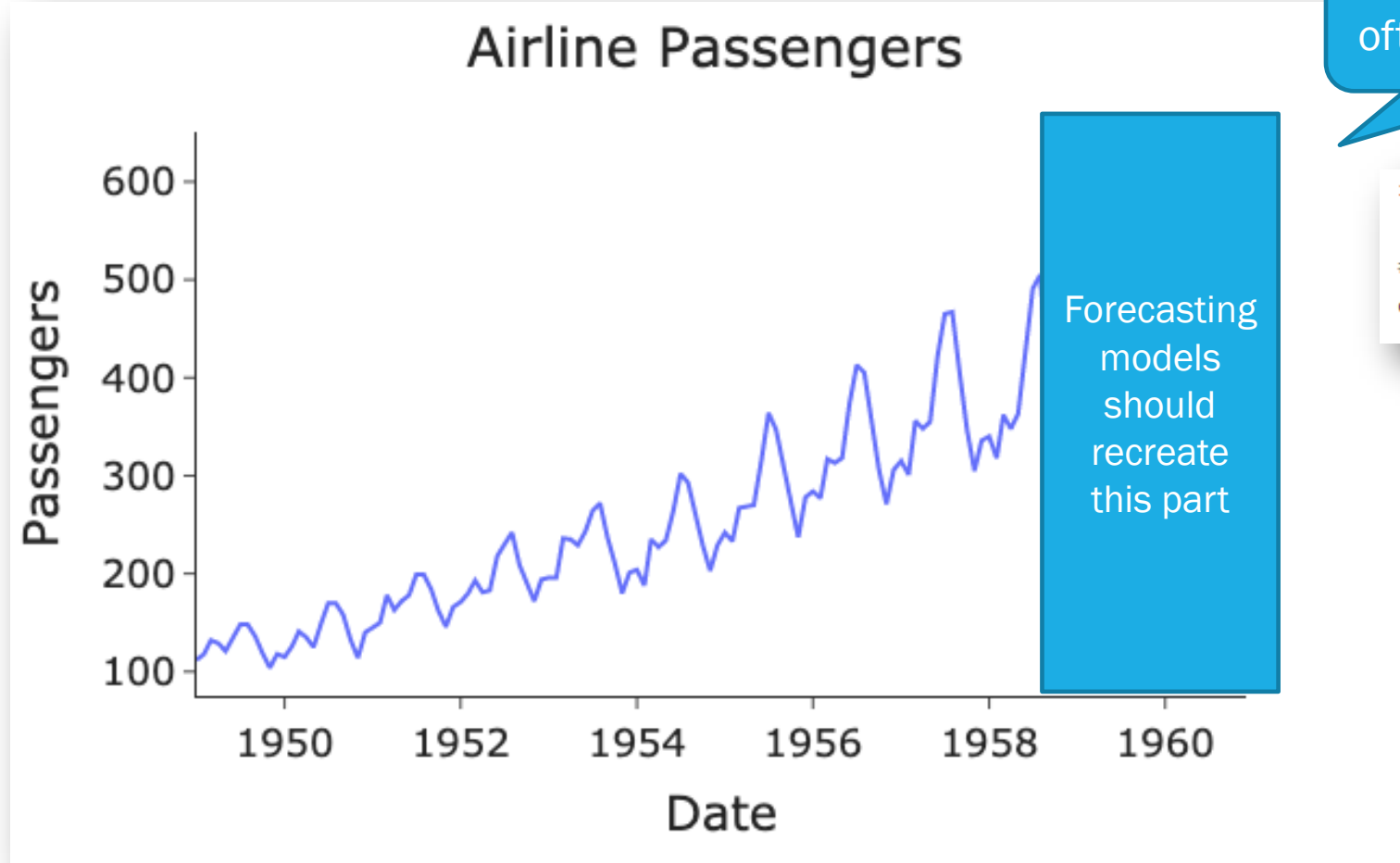


Peter Turner · [Follow](#)

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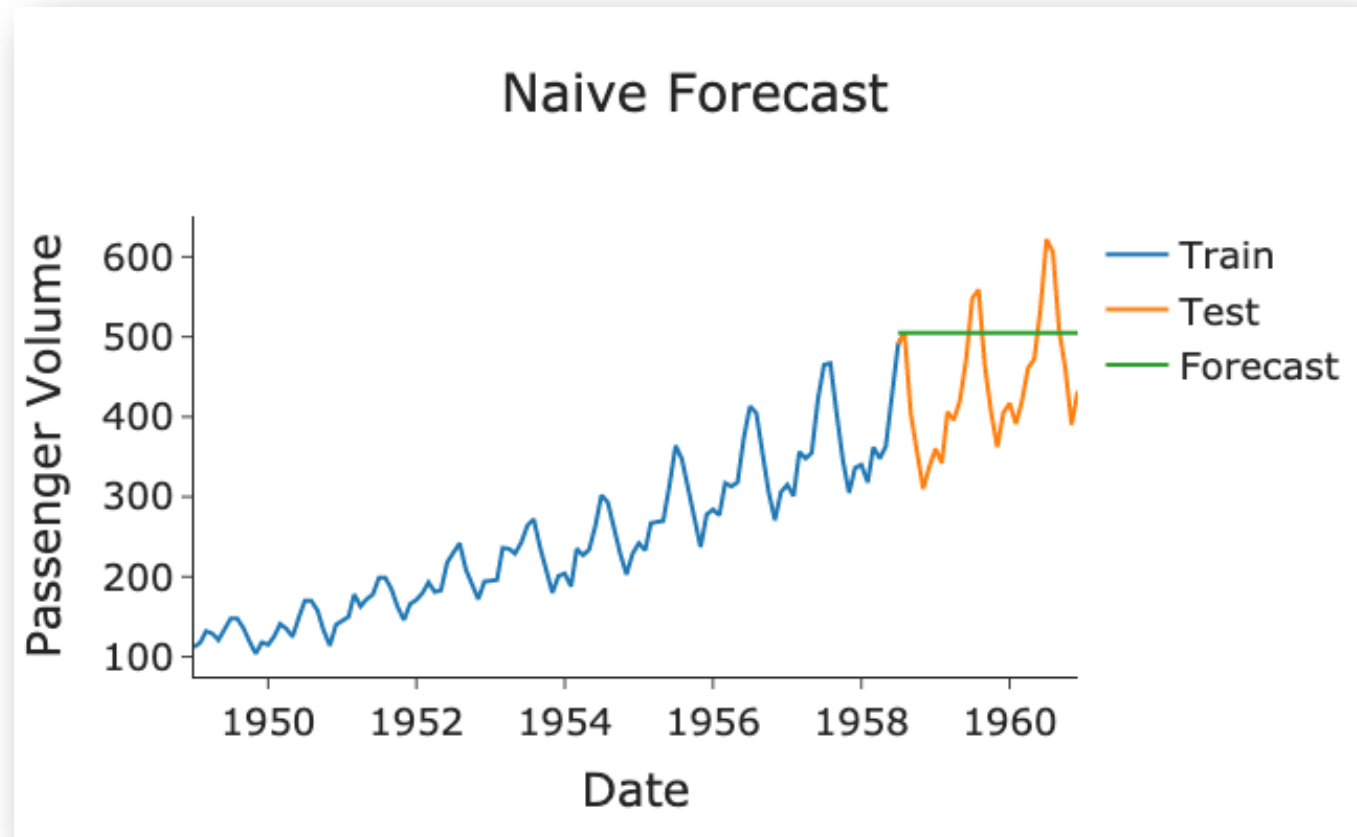
AirPassengers.csv –
often used in tutorials

```
import pandas as pd

# Read in the data
data = pd.read_csv('AirPassengers.csv')
```

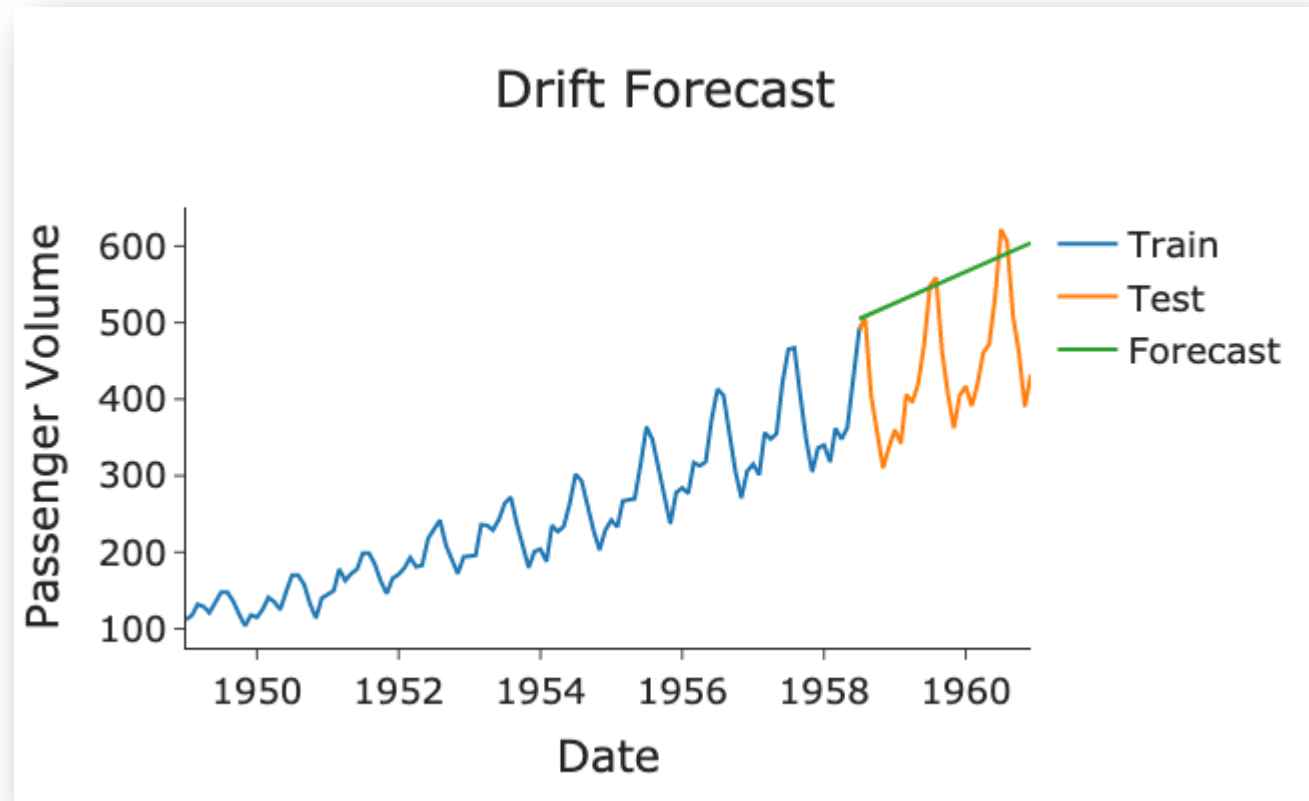
Naive Forecasting

$$y_{T+h} = y_T$$



Drift Model

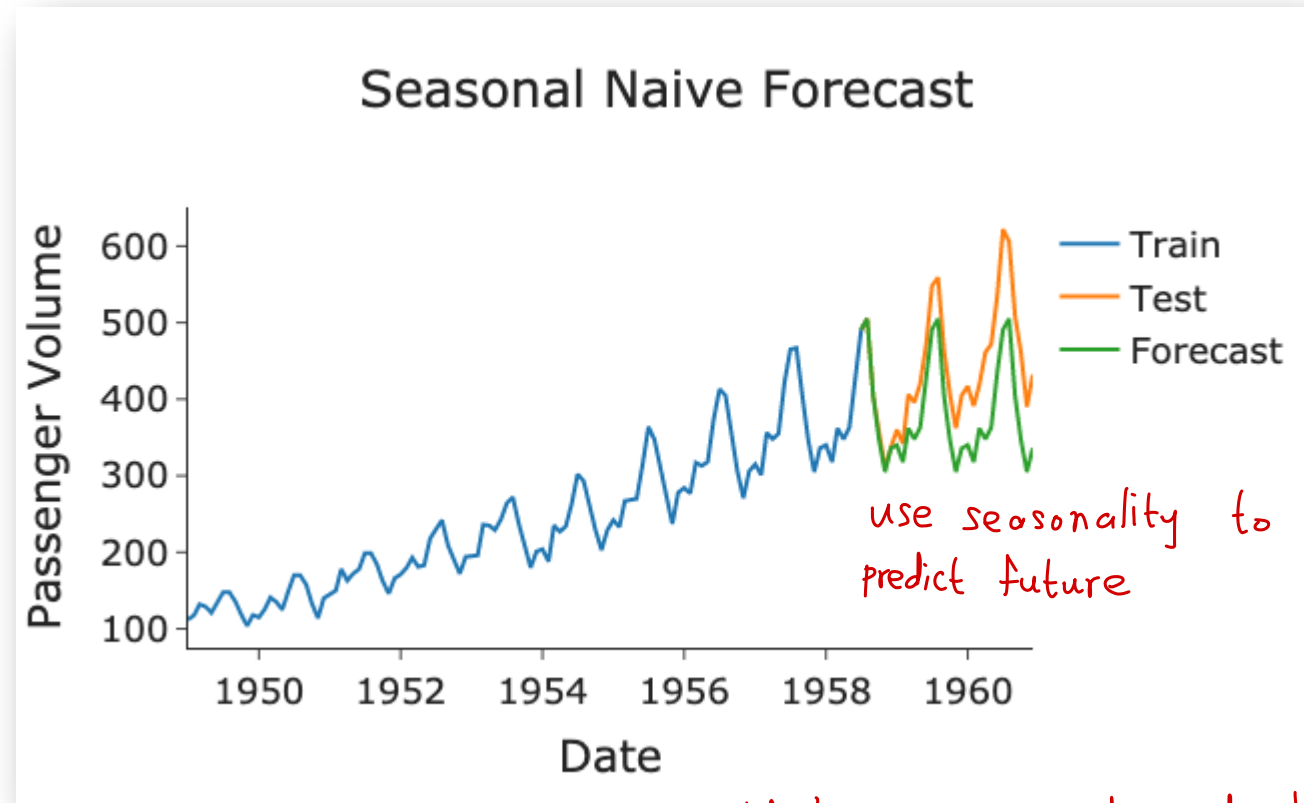
$$y_{T+h} = y_T + h \times \frac{y_T - y_1}{T - 1}$$



Seasonal Naive Forecasting

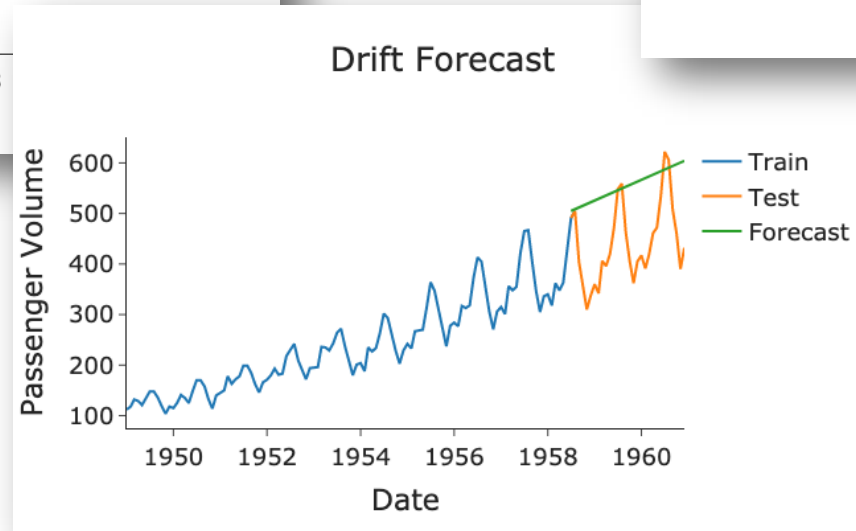
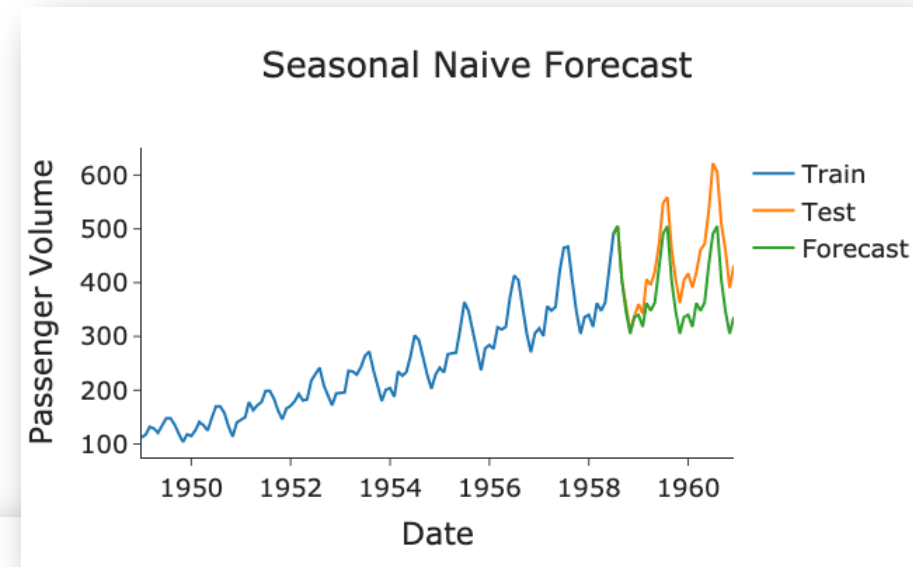
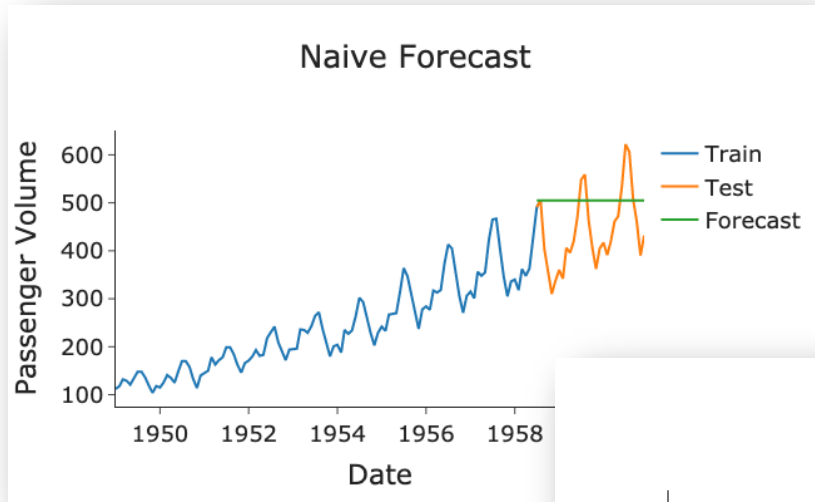
Where m is the seasonality of the data. So, for monthly data with a yearly seasonality $m=12$, quarterly data would have $m=4$ and weekly data would have $m=52$.

$$y_{T+h} = y_{T+h-m}$$



seasonality would be domain dependent

Baseline Models



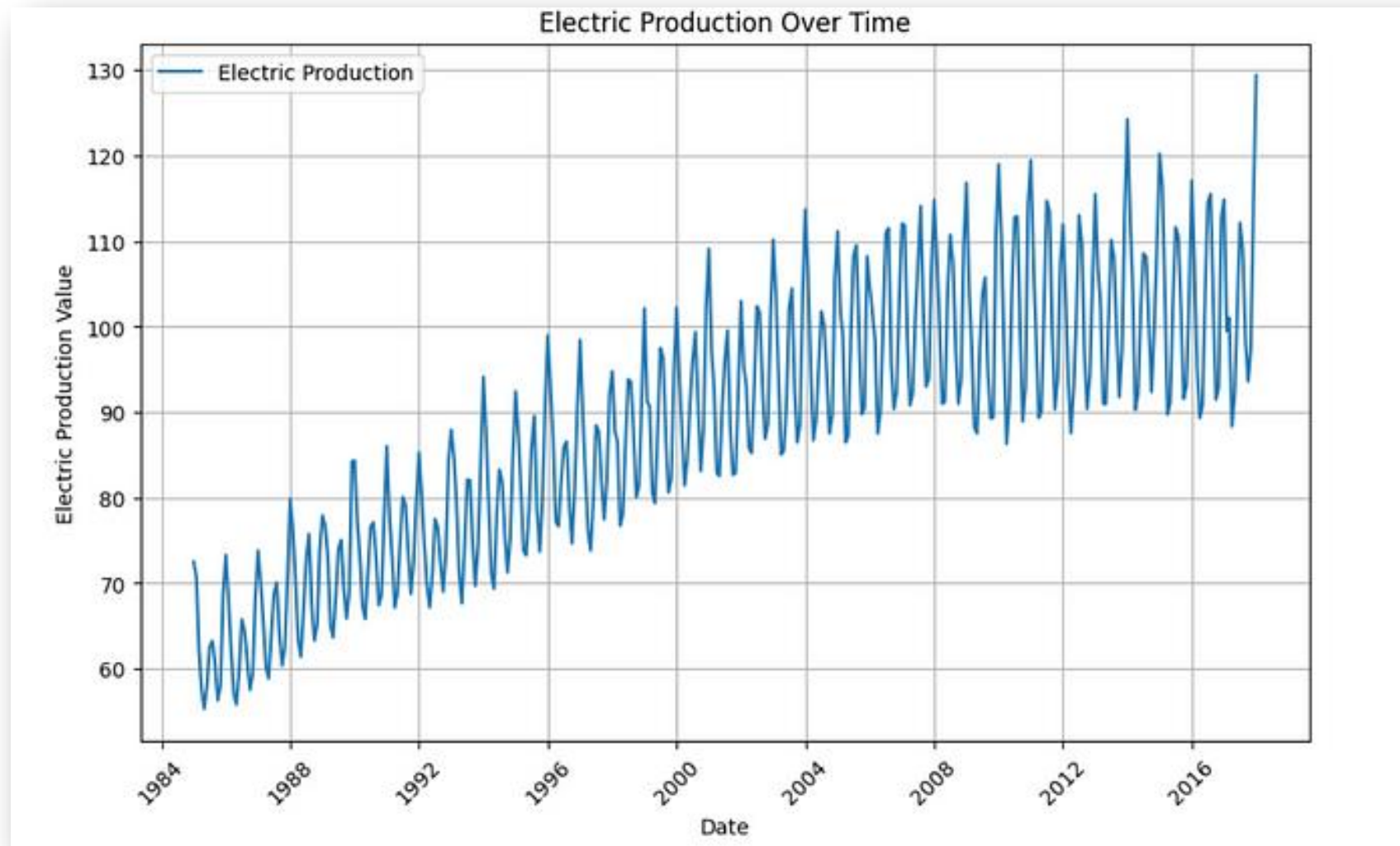
Can we do better?

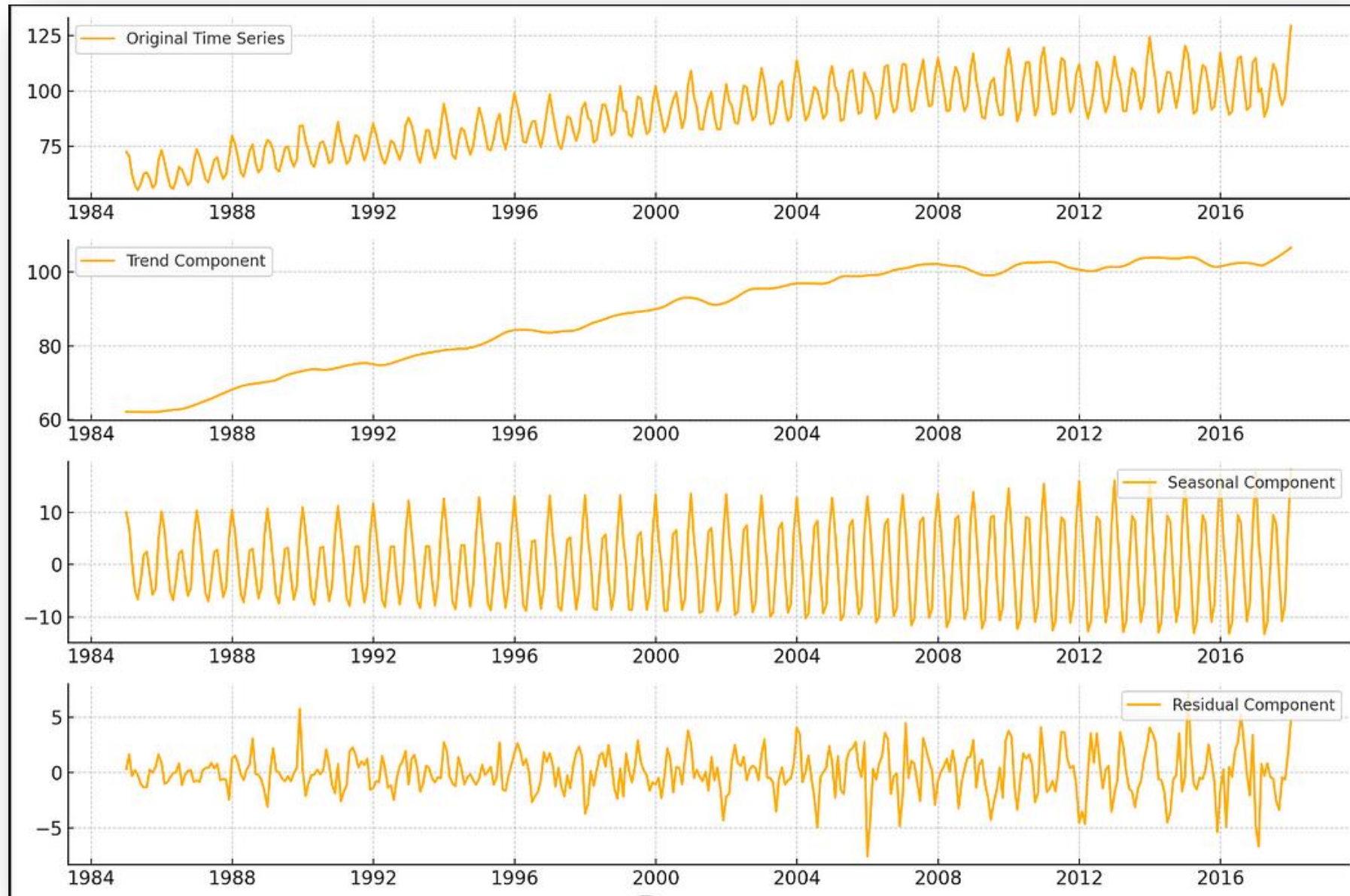
Different models

- ETS models
- ARIMA models
- ML models

ETS modeling

Understanding Error, Trend, Seasonality





Exploring Time Series Data: Unveiling Trends, Seasonality, and Residuals



Venujkenk · [Follow](#)
8 min read · Sep 2, 2023

[Source](#)

Trend

At the heart of every time series dataset lies a trend, a fundamental pattern that reveals the overarching direction of the data. Trends can be linear, indicating steady growth or decline over time, or nonlinear, exhibiting more complex behaviors. **A trend can be upward, downward, or flat.** Identifying trends is vital because they can influence the overall direction of your forecasts.

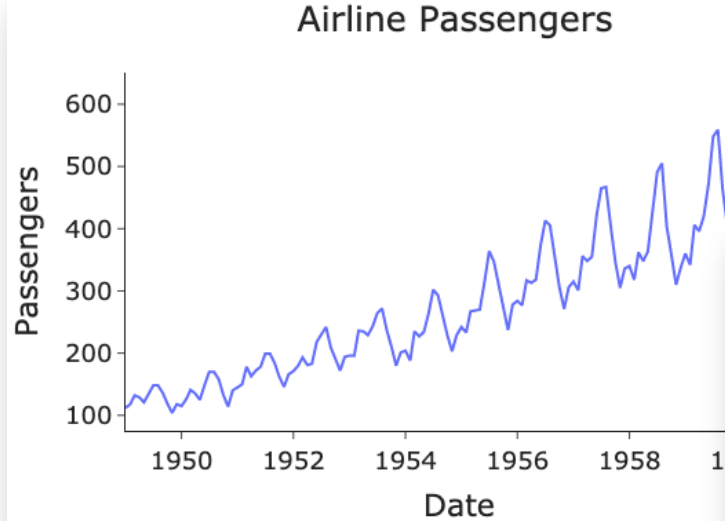
Seasonality

Seasonality refers to **periodic fluctuations** in your data that occur at regular intervals, such as daily, weekly, monthly, or yearly. Think of temperature variations throughout the year. Or, for example, retail sales often peak during the holiday season, while electricity consumption may spike during summer months due to air conditioning use.

Residual

Residuals are the unexplained components of a time series after removing the trend and seasonality. They contain the random noise, irregular fluctuations, and unexpected events that are part of any real-world data.

As simple as « decompose »



```
result = seasonal_decompose(df['#Passengers'], model='additive')

fig, (ax1, ax2, ax3, ax4) = plt.subplots(4, 1, figsize=(10, 12))

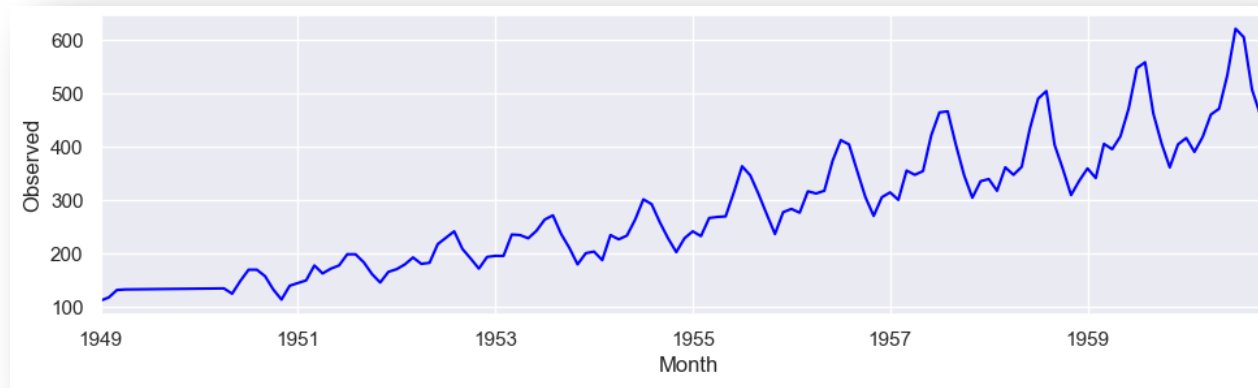
result.observed.plot(ax=ax1, color="blue")
ax1.set_ylabel('Observed')

result.trend.plot(ax=ax2, color='blue')
ax2.set_ylabel('Trend')

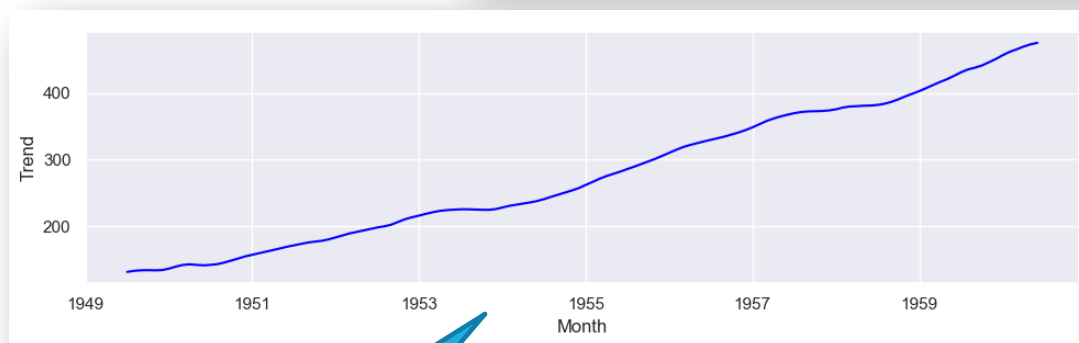
result.seasonal.plot(ax=ax3, color='blue')
ax3.set_ylabel('Seasonal')

result.resid.plot(ax=ax4, color='blue')
ax4.set_ylabel('Residual')

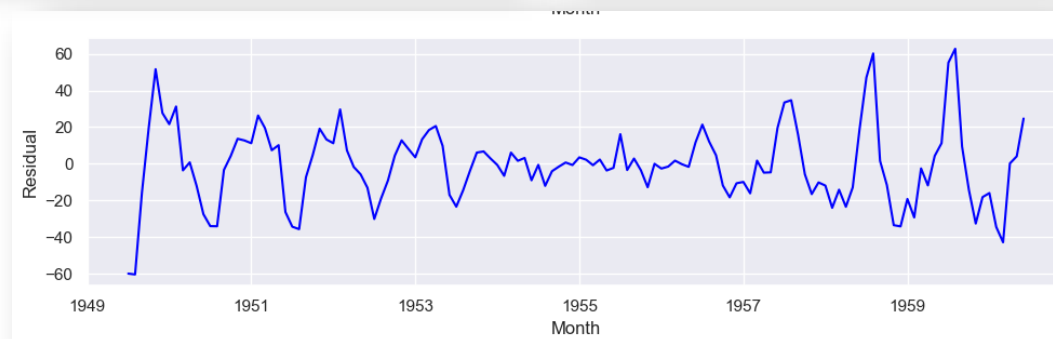
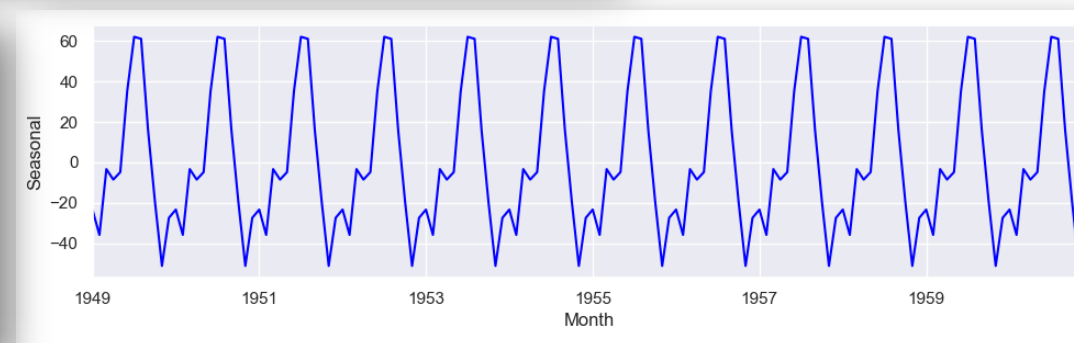
plt.tight_layout()
plt.show()
```



Rule-based or
polynomial
approximation



Rolling average or
linear regression



ETS Model

Additive vs Multiplicative Model

$$Y = T + S + R$$

$$Y = T \times S \times R$$

Where Y is the series, T is the trend, S is the seasonality and R is the residual component.

It is possible to convert a multiplicative model to an additive one by simply taking the log transform.

$$\ln(Y) = \ln(T) + \ln(S) + \ln(R)$$

Simple Exponential Smoothing

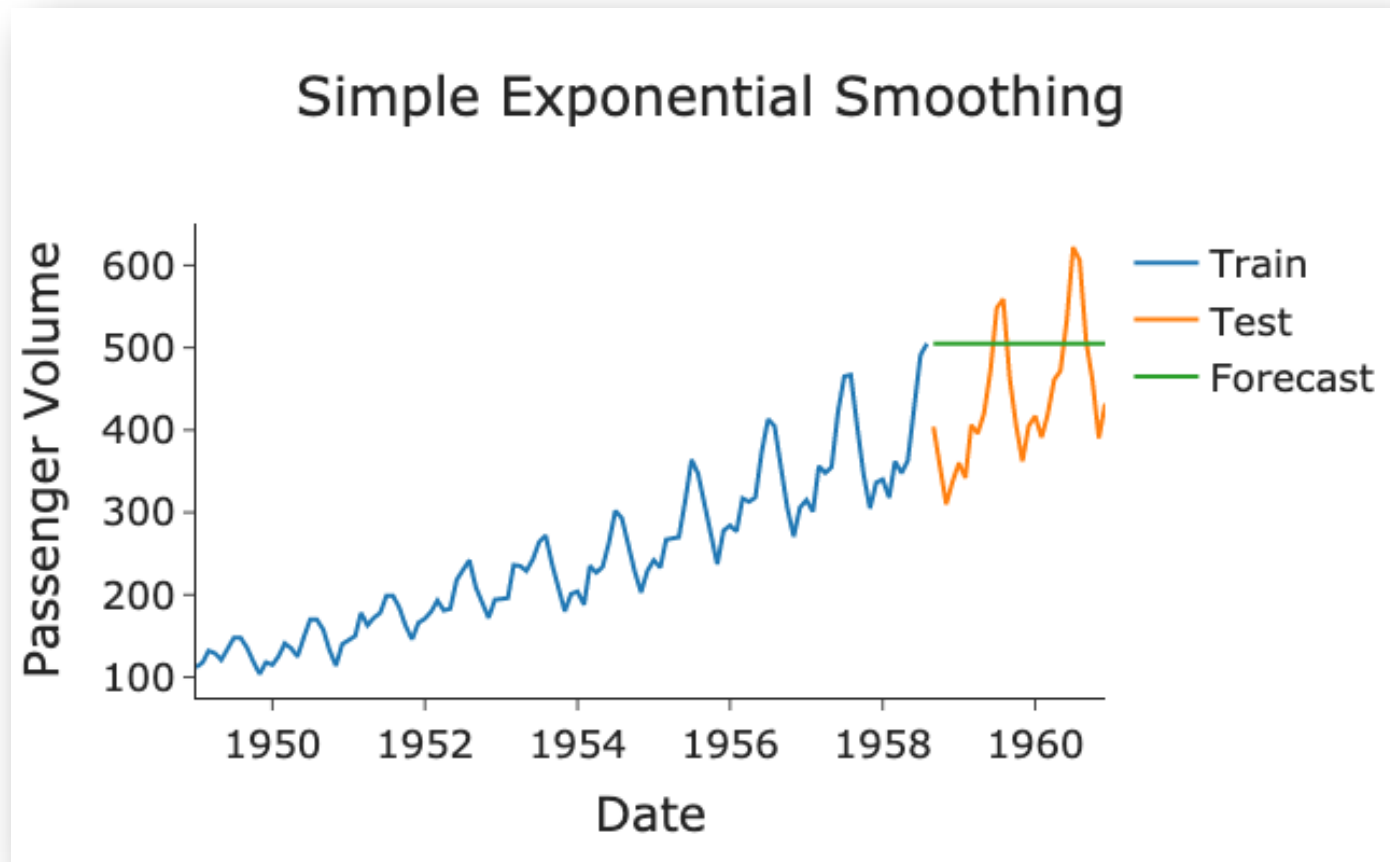
A forecast that is simply a weighted average of the previously observed values.

$$\hat{y}_{T+1|T} = \alpha y_T + \alpha(1 - \alpha)y_{T-1} + \alpha(1 - \alpha)^2 y_{T-2} + \dots,$$

Rewrite for h as any future time step ,and introducing the notion of “level”.

$$\text{Overall Equation : } \hat{y}_{t+h} = l_t$$

$$\text{Level Equation : } l_t = \alpha y_t + (1 - \alpha)l_{t-1}$$



Holt Linear Trend Exponential Smoothing

Charles Holt extended Simple Exponential Smoothing to include a trend component (1957).

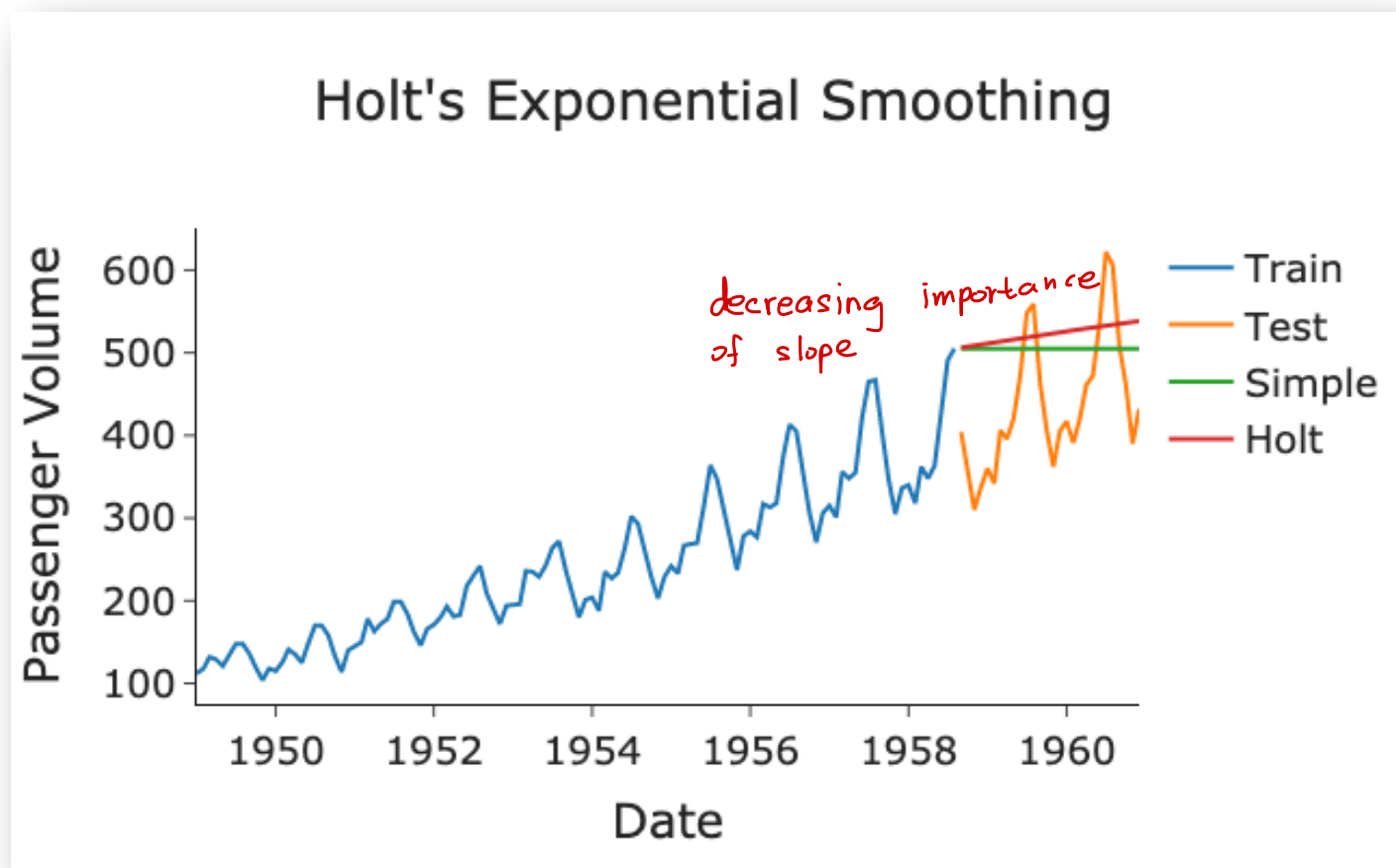
$$\text{Overall Equation : } \hat{y}_{t+h} = l_t + hb_t$$

Use a forecast trend component

$$\text{Level Equation : } l_t = \alpha y_t + (1 - \alpha)(l_{t-1} + b_{t-1})$$

$$\text{Trend Equation : } b_t = \beta(l_t - l_{t-1}) + (1 - \beta)b_{t-1}$$

Smoothing trend



Holt Winters Exponential Smoothing

Adding Trend (Holt)

$$\text{Overall Equation : } \hat{y}_{t+h} = l_t + hb_t + s_{t+h-m}$$

Previous « level »
(baseline model)

Level

Trend

Further adding seasonality
(Winters)

Seasonality

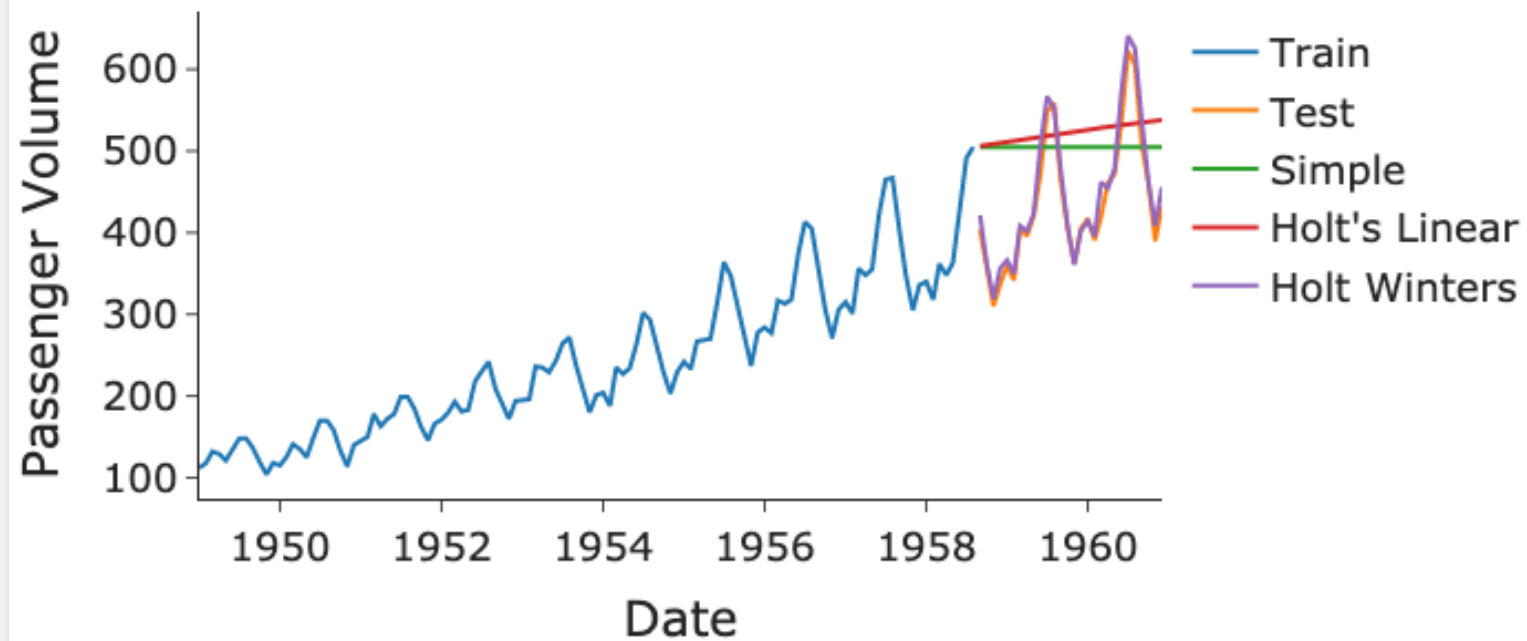
Note how each term
has its own smoothing

$$\text{Level Equation : } l_t = \alpha(y_t - s_{t-m}) + (1 - \alpha)(l_{t-1} + b_{t-1})$$

$$\text{Trend Equation : } b_t = \beta(l_t - l_{t-1}) + (1 - \beta)b_{t-1}$$

$$\text{Seasonality Equation : } s_t = \gamma(y_t - l_{t-1} - b_{t-1}) + (1 - \gamma)s_{t-m}$$

Holt-Winters Exponential Smoothing



This model is still quite used today for forecasting (and available in different libraries).

Evaluation

Where:

- A_t = Actual value at time t
- F_t = Forecasted value at time t
- n = Number of data points

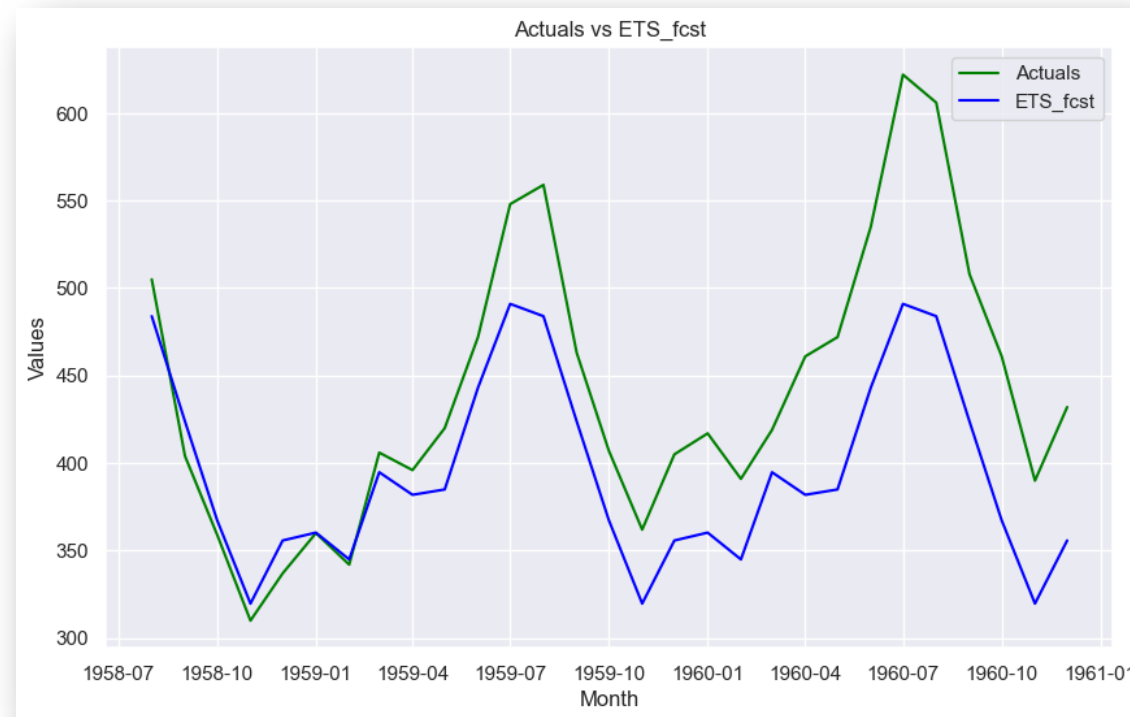
Mean Squared Error (MSE)

$$\text{MSE} = \frac{1}{n} \sum_{t=1}^n (A_t - F_t)^2$$

Mean Absolute Error (MAE)

$$\text{MAE} = \frac{1}{n} \sum_{t=1}^n |A_t - F_t|$$

Known is the time series
before this



Resources

DATA SCIENCE

Basic Time Series Forecasting Techniques

An overview of some very simple forecasting models

Egor Howell
Dec 2, 2022 • 5 min read

[Source](#)

ARTIFICIAL INTELLIGENCE

Time Series Forecasting with Simple Exponential Smoothing

An introduction to exponential smoothing and implementing it in Python to carry out forecasts.

Egor Howell
Dec 9, 2022 • 5 min read

[Source](#)

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Time Series Forecasting with Holt's Linear Trend Exponential Smoothing

Compensating for trend in exponential smoothing models

Egor Howell
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Series of articles by Egor Howell

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Time Series Forecasting with Holt Winters'

A discussion and implementation of the most powerful and useful exponential smoothing model

Egor Howell
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Seasonality in Time Series Analysis

An intuition of how seasonality affects time series analysis

Egor Howell
Oct 26, 2022 • 4 min read

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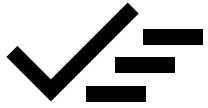
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Time Series Decomposition

Breaking down a time series into its fundamental building blocks

Egor Howell
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TIME SERIES ANALYSIS

- Definition and Tasks
- Segmentation
- Classification
- Forecasting